



Lightweight data security system for resource-constrained IIoT devices[☆]

Chimeremma Sandra Amadi^a, Simeon Okechukwu Ajakwe^b,^{*}
Chigozie Athanasius Nnadiokwe^a, Jun Taesoo^a

^a IT Convergence Engineering Department Kumoh National Institute of Technology, South Korea

^b Smart Computing Department Kyungdong University, Global Campus, South Korea

ARTICLE INFO

Keywords:

Big data
Blockchain
Cyber attacks
Intrusion detection system
Federated learning
Lightweight
Industrial IoT
Security
Zero-day attacks

ABSTRACT

The rapid expansion of the Industrial Internet of Things (IIoT) introduces significant security challenges, particularly for devices with limited computation, memory, and energy. This paper presents a lightweight federated learning (FL) intrusion detection framework integrated with a resource-efficient blockchain layer called PureChain. A trimmed 1D CNN-LSTM model is designed to minimize complexity while maintaining high detection performance, requiring only 41,762 parameters and a 0.17 MB footprint. Experiments on the CIC-Collections2023 dataset and an internally generated IIoT dataset demonstrate strong generalization with accuracies of 97.7% and 96.84%. Five-fold cross-validation confirms stable learning with low variance, and resource overhead measurements show 3.1-ms inference latency and 0.42-mJ energy cost, confirming suitability for constrained IIoT nodes. PureChain provides tamper-evident model validation with a gas cost of 65,618 gas and achieves a 98% contract security score following formal auditing. A structured threat model outlines adversaries, system assumptions, and mitigation mechanisms. Overall, the proposed architecture offers an efficient, secure, and deployable solution for next-generation IIoT intrusion detection systems.

1. Introduction

The rapid advancement of digital technologies, artificial intelligence (AI), and distributed automation has accelerated the transition toward hyper-connected industrial ecosystems, often described as the Internet of Everything. Within this transformation, the Industrial Internet of Things (IIoT) has emerged as a core enabler for modern manufacturing, process automation, and cyber-physical coordination. IIoT systems interconnect heterogeneous industrial devices, sensors, programmable logic controllers (PLCs), actuators, and edge nodes supporting real-time data collection, autonomous decision-making, and adaptive control. However, this pervasive connectivity significantly expands the attack surface and exposes IIoT infrastructure to sophisticated cyber threats, particularly zero-day attacks that exploit previously unknown vulnerabilities and bypass traditional detection mechanisms [1]. Lightweight IIoT nodes, including embedded sensors and low-power controllers, remain particularly vulnerable due to their limited computational, memory, and energy budgets. These constraints hinder the deployment of conventional deep learning-based intrusion detection systems (IDS). Recent research has explored the complementary integration of federated learning (FL) and blockchain (BC) (FL-BC) as

a promising direction to overcome these limitations [2–4]. FL enables distributed model training without centralizing raw data, enhancing privacy, while blockchain ensures transparency, auditability, and tamper-resistant storage of global models. Despite these promising developments, existing FL-BC IDS solutions continue to face several challenges. Most approaches, succinctly captured in Table 1 introduce substantial computational overhead, are not specifically optimized for lightweight IIoT devices, and often lack end-to-end evaluation across both AI and blockchain layers. Furthermore, many studies do not report blockchain performance metrics, such as gas costs, latency, or smart contract security, making the real-world feasibility difficult to assess. These gaps underscore the need for an efficient, resource-aware, and verifiable FL-BC architecture tailored to constrained IIoT environments.

Beyond IIoT, the combination of FL and blockchain has been widely explored across various cyber-physical domains. For example, FL has been applied in smart grid anomaly detection to support privacy-preserving demand response and secure load forecasting [16]. In remote healthcare/IoMT systems, FL enables tamper-resistant medical sensor authentication and privacy-preserving diagnostic models [17].

[☆] This work was partly supported by IITP Innovative Human Resource Development for Local Intellectualization Program (MSIT) (IITP-2026-RS-2020-II201612, 34%), IITP ITRC Program (MSIT) (IITP-2026-RS-2024-00438430, 33%), NRF Basic Science Research Program (MOE) (2018R1A6A1A03024003, 33%), and RISE-(Regional Growth Innovation LAB) Program through Gyeongbuk RISE Center funded by MOE and Gyeongsangbuk-do, Republic of Korea (2026-rise-15-105).

^{*} Corresponding author.

E-mail address: simeon.ajakwe@kumoh.ac.kr (S.O. Ajakwe).

Table 1

Comparison of the proposed model over related works.

Ref	Decentralized storage	Lightweight blockchain network	Lightweight model	Model evaluation time	Gas cost	Efficiency of global model	Blockchain security audit	Privacy and confidentiality	Lightweight IIoT nodes
[5]	✓	✓	✗	✗	✗	✓	✗	✓	✗
[6]	✓	✓	✓	✗	✗	✓	✗	✓	✗
[7]	✓	✓	✗	✗	✓	✓	✓	✓	✗
[8]	✓	✓	✓	✗	✓	✓	✗	✓	✗
[9]	✓	✗	✗	✗	✗	✓	✗	✓	✗
[10]	✓	✗	✗	✗	✗	✓	✗	✓	✗
[11]	✓	✗	✗	✗	✗	✓	✗	✓	✗
[12]	✓	✗	✗	✗	✗	✓	✗	✓	✗
[13]	✓	✗	✗	✗	✗	✓	✗	✓	✗
[14]	✓	✗	✗	✓	✗	✓	✗	✓	✗
[15]	✓	✗	✗	✗	✗	✓	✗	✓	✓
[Ours]	✓	✓	✓	✓	✓	✓	✓	✓	✓

In autonomous vehicles and intelligent transportation systems (ITS), FL facilitates collaborative cyber-attack detection and threat prediction across heterogeneous vehicular fleets [18]. In smart manufacturing and robotic quality control, FL enables distributed fault detection and cross-factory learning while preserving proprietary data [19]. Similarly, blockchain has been leveraged for trustworthy coordination in several industrial domains. It has been used for secure industrial automation logging [20], supply-chain traceability and provenance verification [21], energy metering and decentralized smart-grid billing [22], and trusted collaborative learning across AIoT environments [23]. These diverse applications highlight the growing importance of FL-BC integration for distributed, security-critical infrastructures and reinforce the need for solutions capable of operating on lightweight devices, an area still underexplored in current FL-BC IDS research.

Recent survey studies on FL-based IDS and blockchain-assisted IIoT security [24,25] identify multiple unresolved issues: (i) inadequate support for lightweight IIoT devices; (ii) absence of complete evaluation across both AI and blockchain components; (iii) lack of blockchain-side metrics such as gas cost, transaction latency, and smart-contract vulnerability analysis; (iv) poor integration of FL and BC into a unified, resource-efficient pipeline suitable for constrained cyber-physical environments. Most existing systems optimize model accuracy while ignoring blockchain feasibility, or focus on blockchain storage without demonstrating suitability for low-power IIoT nodes. Many also rely exclusively on centralized datasets and overlook the heterogeneous, distributed nature of IIoT data streams. These limitations motivate the need for a holistic and lightweight FL-BC framework that is computationally efficient, verifiable, and deployable on constrained industrial devices. This study proposes a lightweight security framework that integrates a trimmed 1D CNN-LSTM FL model with a resource-optimized blockchain (PureChain) to deliver secure, efficient, and tamper-evident intrusion detection for IIoT environments. The main contributions of this work are:

- **Lightweight FL architecture for IIoT:** A computationally efficient 1D CNN-LSTM model specifically designed for low-memory and low-CPU IIoT nodes, enabling privacy-preserving collaborative learning.
- **Resource-optimized blockchain layer (PureChain):** Integration of Proof of Authority and Association (PoA²) and SAM+ mechanisms provides low gas cost, transparent model versioning, and tamper-resistant global model storage.
- **End-to-end evaluation:** Joint assessment of global model accuracy, F1-score, inference cost, blockchain gas consumption (65618 gas), and smart contract security (98%), offering the most complete resource and security profiling among existing FL-BC IDS systems.

- **Deployable IIoT security framework:** Demonstrating 97.7% accuracy and 97.1% F1-score while maintaining low resource overhead, validating feasibility for real-time anomaly detection in industrial settings.

The remainder of this article is structured as follows. Section 2 describes the system architecture and methodology. Section 3 presents the implementation and experimental results. Section 4 provides discussions and practical implications. Section 5 concludes the paper and outlines future research directions.

2. Related works

Various researchers have independently developed compelling IDS utilizing FL specifically for IIoT environments [26–29]. Others have focused solely on leveraging blockchain for IDS [30–33]. Although previous studies have explored the applications of AI and blockchain in IIoT [34–37], they did not specifically address the combination of blockchain and FL for lightweight IIoT systems. More recent research has concentrated on the integration of blockchain and FL in IIoT [3, 8,9,38]. However, this proposed system uniquely targets resource-constrained IIoT systems by adapting and integrating a tailored, lightweight FL and lightweight blockchain system by optimizing the decentralized training models as well as the PureChain decentralized network environment, with a focus on enhancing security, privacy, and resource efficiency. [39] examined the application of machine learning for IDS mitigation using recurrent neural networks (RNNs) specifically within the context of medical IoT. Other researchers, such as [40], implemented deep learning within a fog-based architecture, while [41] focused on cloud-based techniques for detecting zero-day attacks. Similarly, [35] introduced a model that combines blockchain technology with a Lightweight Consensus Proof-of-Work (LCPoW) algorithm and a Convivial Optimized Sprinter Neural Network (COSNN) to enhance AI-based security. Their framework leverages blockchain to preserve data privacy and utilizes AI for attack detection. However, this approach did not incorporate an FL framework, which is contrary to the findings of [42], where it was demonstrated that FL significantly enhances data privacy in lightweight IIoT systems.

FL, introduced by Google, is designed to enhance traditional machine-learning frameworks through collaborative efforts. Several systems have integrated FL into IDS, as highlighted by [2,3]. [4] explored the integration of blockchain with FL for secure data sharing, though their study did not address intrusion detection. In [5], the authors combined federated learning with blockchain but focused on detecting only one type of intrusion attack. Similarly, the FLBC-IDS model, proposed in [6], utilized both efficientNET and convolutional neural network (CNN) models, with a permissioned blockchain, which does not address the trust issues associated with aggregation nodes and falls short of realizing the full decentralized potential of blockchain

Table 2

Comparative analysis with state-of-the-art models under identical conditions.

Ref	Decentralized storage	Lightweight blockchain network	Lightweight model	Accuracy (%)	F1-Score	Recall (%)	Training time	Gas cost	Computational efficiency	Resource overhead
[4]	✓	✗	✗	97.5	95.3	94.8	1200	–	Medium	High
[3]	✓	✓	✓	98.9	96.7	96.3	1150	–	High	High
[5]	✓	✓	✗	99.3	98.0	97.5	1800	63 007	Medium	High
[6]	✓	✓	✓	99.2	98.2	98.0	1600	65 618	High	Medium
[7]	✓	✗	✗	94.0	93.0	92.5	1300	–	Medium	High
Ours	✓	✓	✓	97.7	97.1	97.7	899.3	65 618	High	Low

technology. In another study, [7] introduced BFLIDS, a blockchain-empowered FL-based IDS. This model employs blockchain for securing transaction records, FL for data privacy through local model training, the interplanetary file system (IPFS) for decentralized storage, and MongoDB for efficient data management. Ethereum smart contracts are used to manage backend interactions. However, the inclusion of multiple layers of functionality increases the complexity concerning lightweight systems, also where IPFS shares some characteristics with blockchain technology, it does not function as a blockchain network. [8], proposed a blockchain-aided federated learning (BFL)-based IDS framework using a lightweight model for securing metaverse. The lightweight models used here are two complex deep learning models which include multidimensional CNN and multilayer perception (MLP); this combination will certainly improve the accuracy of the system but may not be a good fit for lightweight nodes. They also implemented a hybrid client selection (HCS) technique and an ERC-20 token-based incentive mechanism to reward selected FL clients, good, but however not relevant in a lightweight resource-constrained environment as this will introduce more complexity to the system.

[9], used FL to build a threat-hunting framework called block hunter to automatically hunt for attacks in blockchain-based IIoT networks. Here they used cluster-based architecture with other machine-learning models in the FL environment. Even though they tried to show blockchain for data security, however results discussion and methodology had no trace of blockchain implementation. The models used were also not lightweight in terms of efficiency. In [11], a blockchain and FL system is used to share a threat detection model as appropriate blockchain metrics of cyber threat intelligence. Lastly, in [10], the authors proposed a cooperative IDS based on FL and blockchain, where blockchain ensures privacy preservation for vehicles and enables training for high-accuracy model aggregation for road units. This system did not show the efficiency of the blockchain and FL in terms of lightweight, models implemented, and gas cost of the blockchain network. To address the issues in traditional and existing lightweight IDS IIoT approaches, the proposed work implements a lightweight FL and blockchain framework that preserves data privacy and is specifically designed for resource-constrained nodes. This framework combines lightweight FL and customized blockchain technology, which is based on trimmed CNN and long-short-term memory (LSTM) for the AI model and a public blockchain network built on a Proof of Authority and Association (PoA²) mechanism.

2.1. Comparative analysis

We compared our approach against several state-of-the-art IDS that employ FL and blockchain-based data security enhancements. The comparative analysis, summarized in Table 1, illustrates that our approach prioritizes both security and resource efficiency, which are critical in Industrial Internet of Things (IIoT) environments characterized by devices with limited computational capacities.

Like many existing solutions, the proposed approach utilizes decentralized storage to enhance data availability and reliability. However, we further advance this by integrating a lightweight blockchain network, which ensures data integrity and resilience against tampering without imposing significant computational overhead. This approach

differentiates the proposed model from others, such as those in [5,9], and [10], which do not employ a lightweight blockchain network, making them less suitable for IIoT scenarios where computational resources are scarce. The design of the proposed model specifically targets the needs of lightweight IIoT nodes. As shown, our approach ensures compatibility with these resource-constrained devices by maintaining a lightweight architecture that does not compromise accuracy or security. This is a notable improvement over models like [13,15], which do not support lightweight IIoT nodes effectively.

Unlike other models that focus primarily on security enhancements, the proposed system also achieves a faster model evaluation time. Only a few models, such as [14], exhibit comparable timeliness, but they lack other critical features like a lightweight blockchain network and comprehensive security auditing. This makes the proposed solution more balanced in achieving both efficiency and security. Additionally, the proposed system minimizes Gas Costs associated with blockchain operations, which is a significant advantage over more costly models such as [7,8]. The proposed model implements robust security measures, including a thorough blockchain security audit and strong privacy and confidentiality guarantees, ensuring that critical infrastructure is well-protected. While other models, such as [7,8], offer similar levels of security, they do not provide the same degree of efficiency or compatibility with low-resource devices. Overall, the results presented in Table 1 demonstrate that the proposed lightweight FL and blockchain system offers a unique combination of security, efficiency, and suitability for resource-constrained environments. By achieving a balance between these metrics, our system stands out among state-of-the-art solutions for IIoT environments, offering superior timeliness and security without compromising on resource efficiency.

2.2. Comparative analysis with state-of-the-art models under identical conditions

Table 2 provides a comprehensive comparison of the proposed model with state-of-the-art intrusion detection systems (IDS) employing federated learning (FL) and blockchain. The table highlights key performance metrics, including accuracy, F1-score, recall, training time, gas cost, computational efficiency, and resource overhead. This comparison demonstrates that while systems such as those proposed in [3,5] achieve marginally higher accuracy and F1-scores, they do so at the expense of higher training times, greater computational demands, and limited suitability for lightweight IIoT environments. In contrast, the proposed system strikes a balance between high performance (accuracy: 97.7%, F1-score: 97.1%) and resource efficiency, making it particularly suitable for resource-constrained IIoT devices. The gas cost of the proposed system at 65618 gas aligns closely with other blockchain-enabled systems such as [6], indicating cost efficiency in blockchain operations. Moreover, the training time of 899.3 s is significantly lower than that of other systems, showcasing the advantages of the lightweight model design in reducing computational demands.

Synthesizing the above studies and recent survey papers on federated learning-based IDS and blockchain-assisted IIoT security, three main knowledge gaps emerge: Limited support for lightweight IIoT nodes. Many existing systems employ complex deep architectures (e.g., EfficientNet, GANs, capsule auto-encoders) or multi-component storage

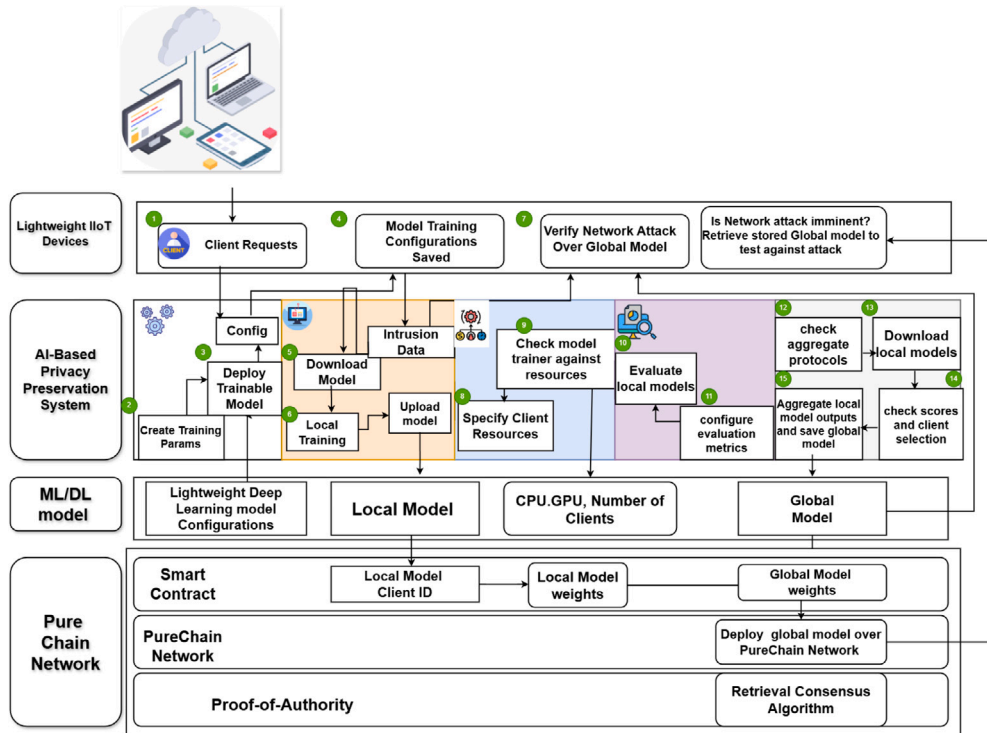


Fig. 1. Architectural framework of the AI-based lightweight blockchain security model, given a trimmed neural network model with filtered single hidden structure and LSTM. The final stages show the global model weight storage over the smart contract and deployed over PureChain respectively.

stacks (e.g., IPFS + MongoDB), which increase memory footprint, training time, and resource overhead, making them unsuitable for constrained IIoT devices. Incomplete integration of FL and blockchain performance evaluation. Several works claim to use blockchain for secure aggregation or logging but do not report concrete gas cost, transaction latency, or security audit metrics, making it difficult to assess real deployment feasibility in industrial networks. Insufficient focus on end-to-end lightweight design. Prior approaches typically optimize either the FL model (accuracy, F1-score) or the blockchain layer (decentralized storage), but rarely treat AI + blockchain as a single co-designed pipeline optimized for both detection performance and resource efficiency. Lack of application to truly resource-constrained IIoT contexts. While some methods target general IoT or vehicular networks, few explicitly design and profile their systems for industrial sensors and edge nodes with limited CPU, memory, and energy budgets. Missing security auditing of smart contracts. Even where blockchain is implemented, smart contracts are often not subjected to independent vulnerability analysis, leaving their robustness against malicious exploitation uncertain. The present work directly addresses these gaps by (i) developing a trimmed 1D CNN-LSTM FL model specifically configured for low-complexity IIoT environments, (ii) integrating a PureChain blockchain that is tailored for reduced gas cost and fast Proof of Authority and Association (PoA²)-based consensus, and (iii) reporting both AI-side metrics (accuracy, F1-score, recall, evaluation time) and blockchain-side metrics (gas cost, transaction cost, and a 98% smart contract security audit score) under comparable conditions. This positions the proposed framework as a lightweight, end-to-end evaluated alternative to existing FL-blockchain IDS schemes.

2.2.1. Cost-benefit interpretation and performance-cost trade-off

Tables 1 and 2 demonstrate that improvements in detection accuracy beyond a certain threshold incur disproportionately higher system costs in blockchain-enabled IIoT IDS deployments. While several state-of-the-art models report marginally higher accuracy values (98.9–99.3%), these gains are accompanied by substantial increases

in model size, training time, energy consumption, and blockchain overhead, which directly conflict with the operational constraints of lightweight IIoT devices. Specifically, models such as MiTFed and FLBC-IDS achieve accuracy improvements of approximately 1.5–1.6% relative to the proposed approach, but require nearly double the training time (1600–1800s versus 899.3s), significantly larger parameter counts, and medium-to-high resource overhead (Table 2). This translates into increased memory usage, higher inference latency, elevated energy consumption per prediction, and more frequent or costly blockchain interactions, all of which degrade deployability in resource-constrained environments.

By contrast, the proposed CNN-LSTM model achieves 97.7% accuracy and a 97.1 F1-score while operating with only 41,762 parameters, millisecond-scale inference latency, and low energy consumption (0.42 mJ per inference). Although this represents a modest absolute accuracy reduction of less than 1.5% compared with heavier baselines, it yields substantial reductions in computational cost (3–7× lower latency), memory footprint (over 30× fewer parameters), and training time. From a system-level perspective, this trade-off results in a higher detection-efficiency-per-cost ratio, making the proposed solution more suitable for continuous monitoring on battery-powered or low-power IIoT nodes. Furthermore, when blockchain costs are considered, the proposed architecture maintains a competitive gas cost (65,618 gas) while uniquely combining lightweight model execution, low latency, and audited smart-contract security. In contrast, higher-accuracy baselines either omit blockchain cost analysis or incur comparable or higher gas usage without achieving similar reductions in computational overhead. Overall, these results indicate that the proposed system operates near an optimal balance point, where further accuracy gains would impose disproportionate increases in memory, latency, energy, and blockchain cost. This balance is well aligned with real-world IIoT deployment requirements, where sustained efficiency, scalability, and reliability are more critical than marginal improvements in classification accuracy.

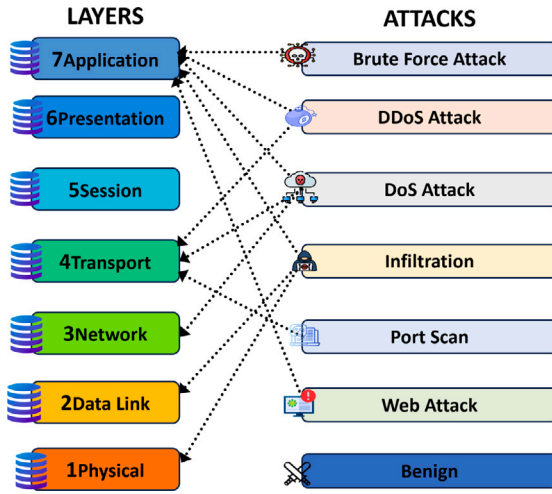


Fig. 2. CIC-Collection dataset is a super fusion of 4 datasets; CIC-IDS2017, CIC-DoS2017, CSE-CIC-IDS2018 and CIC-DDoS2019 with varied attack classes.

3. System design and methodology

This section outlines the overall system design, as illustrated in Fig. 1. The proposed system is a lightweight IIoT-based IDS designed for resource-constrained IoT nodes. The system component comprises interconnected resource-constrained IIoT devices, the privacy preservation unit, the ML/DL cognitive engine for intrusion predictive analysis and pattern discovery, and the PureChain module for security auditing. The first layer of the system consists of lightweight IIoT devices, also referred to as edge devices or clients. These devices operate with constrained resources, such as limited computational power, memory, and energy. Each device registers interest in training its local data d_q on a lightweight neural network L_q . The training process begins when the devices initialize the $get_model()$ function to download a pre-configured model from the central server F_s . After training locally on their data, each edge device generates an updated local model UL_q and securely sends it back to the central server. This decentralized training process reduces the computational burden on individual devices while ensuring that data remains local, thereby enhancing privacy. The second layer, which is the privacy preservation module, ensures the integrity and security of the FL process. It leverages secure protocols to handle the exchange of local models and protects against network intrusions during training. This layer configures training parameters and enforces privacy-preserving mechanisms, such as secure aggregation, to ensure that individual device data cannot be reconstructed from the shared updates. Before sending the models back to the server, the module performs intrusion detection and verifies the integrity of the data being transmitted. This ensures that malicious attacks are mitigated early in the training cycle.

The third layer focuses on the lightweight ML/DL cognitive models used in the FL process. These models are designed to be computationally efficient, making them suitable for lightweight devices. The central server F_s aggregates the local models UL_q received from all participating devices using the federated averaging (FedAvg) algorithm to compute the global model G_q . The global model is then evaluated using pre-configured metrics to ensure optimal performance before being shared back with edge devices. This iterative process of local training and global aggregation ensures that the model continuously improves while leveraging data from multiple devices.

The PureChain module integrates blockchain technology to provide a tamper-resistant and transparent mechanism for recording and managing model updates. Once the global model G_q is computed by the central server F_s , it interacts with the PureChain network via

a smart contract. The smart contract ensures secure storage of the aggregated model on the blockchain. Using the *RegisterCentralServer()* function, the system verifies the authenticity of the central server and prevents unauthorized modifications to the stored model parameters. Subsequently, the *SetModel()* function stores the global model G_q and its learning parameters M as a block in the blockchain. This ensures immutability and transparency. Devices that wish to use the optimized global model for early detection can retrieve it securely using the *GetModel()* function. Additionally, the blockchain provides a retrieval consensus algorithm, which validates the global model before making it available for deployment on edge devices. This system design ensures seamless interaction between its components. The lightweight IIoT devices initiate local training and interact with the privacy preservation module to ensure secure data exchange. The trained models are then aggregated in the ML/DL cognitive model layer, where the global model is optimized. Finally, the PureChain network module securely stores the global model, making it accessible to participating devices while maintaining transparency and tamper-resistance.

3.1. Dataset characterization and preprocessing

3.1.1. CIC-collection2023 dataset

The CIC-Collection2023 dataset is a refined compilation of the CIC-IDS2017, CIC-DoS2017, CSE-CIC-IDS2018, and CIC-DDoS2019 datasets. It was specifically developed for training predictive models to detect network traffic intrusions. This publicly accessible evaluation dataset on Kaggle is commonly used for testing intrusion detection systems. It was selected due to its accurate representation of contemporary network intrusions and its inclusion of a wide variety of real-world attack techniques. With its extensive array of features, this collection aims to provide a larger, more diverse set of NIDS samples for more comprehensive analyses. Although CIC-Collection2023 is not an IIoT-specific dataset, its traffic characteristics closely resemble those observed in industrial networks. Prior IIoT security studies [6,11] have similarly adopted CIC-based datasets due to their richness, the variety of modern attack variants, and their suitability for modeling temporal statistical flow features. Hence, CIC-Collection2023 serves as a representative high-fidelity proxy for evaluating IDS solutions intended for IIoT environments. As shown in Table 3, each CIC-Collections2023 attack vector induces statistical deviations in packet-level features that closely mirror real IIoT adversarial behaviors, including PLC flooding, sensor probing, unauthorized firmware injection, and credential attacks. Mathematically, the perturbation patterns $\delta_A(t)$ align with known IIoT threat signatures, confirming that the dataset provides a realistic approximation of operational IIoT attack dynamics.

The dataset includes 7 distinct types of attack classes, each representing different categories and layers of attacks: DDoS, DoS, BotNet, web attacks, infiltration, port scans, and brute force attacks. These are captured through 57 preprocessed features that effectively mirror the network traffic seen in real-world environments. The dataset comprises 9,167,581 rows and 57 features [43]. Fig. 2 illustrates the classification of attacks within the CIC-Collection 2023 dataset. In this dataset, one-hot encoding was applied to each level of a categorical feature, where a new variable was created, and each category was represented by a binary variable with values of either 0 or 1. Here, 0 indicates the absence, and 1 indicates the presence of that category [44].

To better understand feature relationships in the CIC-Collections2023 dataset, a correlation heatmap of selected numerical attributes is shown in Fig. 3. The strong positive and negative correlations observed between traffic-volume, flow-duration, and header-based statistics provide additional validation of the dataset structure and help motivate the feature selection decisions used in our IDS.

Table 3
Mapping of CIC-Collections2023 attack vectors to real IIoT threat scenarios.

CIC attack vector	Traffic-level behavior ($\delta_A(t)$)	Corresponding real IIoT scenario
DoS/DDoS	$\delta_A(t) = \Delta \text{flow_pkt_rate} \gg \mu_{\text{normal}}$ (high-rate flooding)	PLC or RTU flooding, sensor jamming, gateway saturation, actuator command overload
PortScan	$\delta_A(t) = \text{unique ports accessed} \gg \text{baseline}$ (aggressive probing)	Enumeration of PLC ports/protocols (Modbus, DNP3, Profinet), reconnaissance before intrusion
BruteForce (SSH/FTP)	$\delta_A(t) = \Delta \text{failed_auth} \gg 1$ (credential abuse)	Password-guessing against MQTT brokers, OPC-UA servers, edge controller login portals
Infiltration/Malware Traffic	Increase in payload entropy and irregular flow sizes	Untrusted firmware upload, remote code injection into PLCs, propagation of IIoT botnets
Botnet/C+C Traffic	Periodic beaconing patterns and low-entropy command packets	Compromised IIoT cameras, HVAC units, or industrial sensors participating in coordinated attacks
Web Attacks/Recon Traffic	Abnormal TLS handshake sequences	Exploitation attempts on encrypted IIoT gateways or VPN concentrators

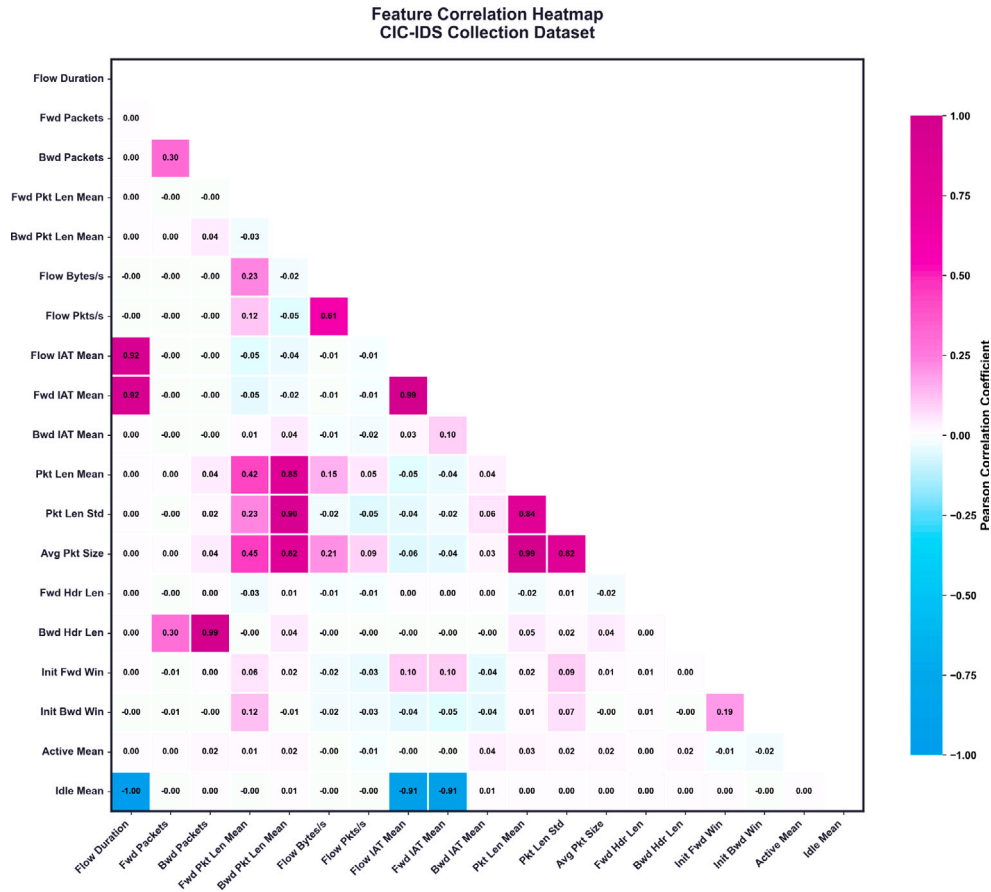


Fig. 3. Correlation heatmap of key numerical features from the CIC-Collections2023 dataset, highlighting positive and negative dependencies relevant for intrusion detection.

3.1.2. Internal lightweight IIoT dataset

To complement the publicly available CIC-Collections2023 dataset and to better reflect deployment conditions in constrained industrial environments, we constructed a synthetically generated internal lightweight IIoT dataset. The purpose of this dataset is to evaluate the proposed federated IDS under controlled, IIoT-specific conditions that more closely resemble small-scale industrial networks composed of sensors, programmable logic controllers (PLCs), and edge gateways. The internal dataset contains 5000 traffic samples with 57 features and a single categorical label (ClassLabel). To maintain architectural compatibility and fair comparison with CIC-Collections2023, the feature space mirrors common flow-level representations used in industrial IDS studies. Specifically, the features are organized into 40 continuous

attributes capturing traffic rates and timing statistics, 10 discrete attributes representing protocol and connection-state indicators, and 7 binary attributes encoding boolean security conditions. Traffic samples were synthetically generated to emulate realistic IIoT device behaviors and attack patterns. Benign traffic represents periodic sensor reporting and control signaling with moderate packet rates and low variance. Attack traffic was generated to reflect three common IIoT threat classes: DoS traffic exhibiting elevated packet and byte rates with reduced inter-arrival times; PortScan traffic characterized by sparse connections and systematic probing behavior; and BruteForce traffic modeling repeated authentication attempts with distinctive failure indicators. Class proportions were intentionally imbalanced, with approximately 50% of samples normal traffic and the remaining samples distributed across the three attack classes, reflecting realistic operational conditions in industrial networks.

Algorithm 1: Synthetic Lightweight IIoT Dataset Generation

Input: Number of samples $N = 5000$, number of features $F = 57$
Output: $X \in \mathbb{R}^{N \times F}$, labels y

- 1 Set random seed to 42;
- 2 Define class labels: 0 =Normal, 1 =DoS, 2 =PortScan, 3 =BruteForce;
- 3 Define class proportions $p = [0.50, 0.20, 0.15, 0.15]$;
- 4 Sample y from categorical(p);
- 5 **for** $i \leftarrow 1$ **to** N **do**
- 6 **if** $y_i = 0$ (Normal) **then**
- 7 Draw 40 continuous features from $\mathcal{N}(0, 1)$;
- 8 Draw 10 integer features from $\{0, \dots, 5\}$;
- 9 Draw 7 binary flags from Bernoulli(0.1);
- 10 **else**
- 11 **if** $y_i = 1$ (DoS) **then**
- 12 Draw 40 continuous features from $\mathcal{N}(1.5, 1.2)$;
- 13 Draw 10 integer features from $\{1, \dots, 7\}$;
- 14 Draw 7 binary flags from Bernoulli(0.4);
- 15 **else**
- 16 **if** $y_i = 2$ (PortScan) **then**
- 17 Draw 40 continuous features from $\mathcal{N}(0.5, 0.8)$;
- 18 Draw 10 integer features from $\{0, \dots, 10\}$;
- 19 Draw 7 binary flags from Bernoulli(0.3);
- 20 **else**
- 21 // $y_i = 3$ (BruteForce)
- 22 Draw 40 continuous features from $\mathcal{N}(0.8, 1.0)$;
- 23 Draw 10 integer features from $\{0, \dots, 8\}$;
- 24 Draw 7 binary flags from Bernoulli(0.5);
- 25 **end**
- 26 **end**
- 27 Concatenate the 57 features to form row X_i ;
- 28 **end**
- 29 Export (X, y) to CSV as `internal_iiot_dataset.csv`;

The dataset generation process follows explicitly defined statistical distributions for each traffic class and feature group, with a fixed pseudorandom seed to ensure reproducibility. The full generation procedure is summarized in Algorithm 1, and the corresponding Python script used to produce the dataset is publicly available [45]. Compared with CIC-Collections2023, which contains millions of samples spanning diverse enterprise and network environments, the internal IIoT dataset is smaller but more focused. While CIC-Collections2023 offers greater diversity and complexity, the internal dataset provides a controlled representation of IIoT-specific traffic patterns, reduced feature variance, and device-level behavior that are typical of resource-constrained industrial deployments. Evaluating the proposed model on both datasets therefore enables assessment of scalability on large public benchmarks, as well as robustness and generalizability under realistic IIoT operating conditions.

3.2. Trimmed FL strategy and model architecture

To achieve a lightweight and privacy-preserving federated learning (FL) workflow suitable for IIoT devices, we employ a trimmed 1D CNN–LSTM architecture designed to minimize memory usage and computational overhead. The selection of a hybrid 1D CNN–LSTM model is motivated by the intrinsic structure of IIoT network traffic, which consists of temporally ordered packet-flow statistics exhibiting both short-range correlations (e.g., burst rates, protocol flags) and longer-term dependencies arising from periodic sensing, control loops, and sustained attack behaviors. Pure CNN-based models effectively capture local statistical patterns but lack explicit temporal memory, while standalone recurrent architectures incur higher computational cost when applied directly to high-dimensional traffic features. The proposed hybrid design decomposes the learning task by using lightweight convolutional layers as a feature compressor, followed by an LSTM layer that models temporal dependencies over the reduced representation,

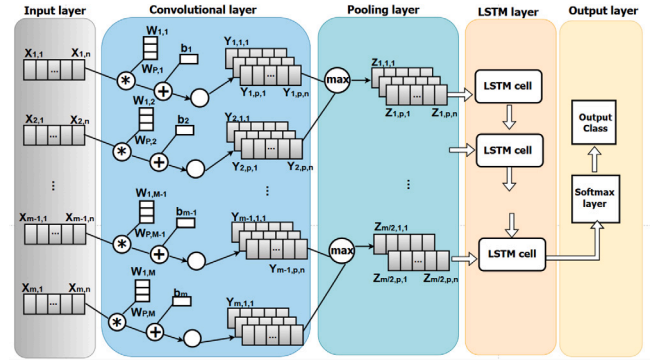


Fig. 4. Configuration of the proposed model structure using a one-dimensional (1-D) CNN and LSTM recurrent neural network.

yielding an optimal accuracy/efficiency trade-off for federated IIoT intrusion detection. 1D CNN component extracts temporal statistical patterns from packet-flow sequences using one-dimensional kernels, as illustrated in Fig. 4. Assuming an input vector $x \in \mathbb{R}^{n \times 1}$, where n denotes the sequence length, the convolutional output is given by

$$y^{j,i,k} = f \left[\sum_{i=1}^{\infty} X_{i,k} * W_{j,i} + b_i \right], \quad (1)$$

where $1 \leq i \leq m$ indexes samples, $1 \leq j \leq p$ indexes kernel positions, $1 \leq k \leq n$ indexes feature channels, f denotes the ReLU activation, $*$ is the convolution operator, W and X are the filter weights, and b_i is the bias term. The CNN outputs are passed to a single-layer LSTM unit, which captures sequential dependencies and reduces temporal variance, followed by a fully connected softmax classifier for multi-class intrusion detection. The overall workflow is summarized in

Algorithm 2: Lightweight Global Model Training Process

```

Data: Edge – device data
Result: optimized – model
1 START Server
2 Go to Server Terminal envs
3 INSTALL correct versions of packages: tensorflow,flwr

    DEFINE constants:
        NUM_CLIENTS = 5
        BATCH_SIZE = 32
        NUM_CLASSES = 8
        FEATURES = 57
        DATA_SIZE = 9167581

while True do
    LOAD dataset from Folder
    ENCODE ClassLabels column using LabelEncoder
    SPLIT dataset into NUM_CLIENTS partitions

    for each client partition do
        SPLIT partition into training and testing sets
        DEFINE create_model function:
            CREATE Keras Sequential model
            ADD Layers
            1-d cnn
            lstm
            RETURN model
        INITIALIZE federated learning state
    end

    for each round in NUM_ROUNDS do
        EXECUTE one round of federated learning
        PRINT metrics
        EVALUATE final model on test dataset
    end

end
return optimized – model;

```

Algorithm 2. To support reproducibility and provide full transparency of the computational footprint, the complete model configuration and training hyperparameters are presented in Table 4. The architecture consists of two convolutional layers (16 and 32 filters, kernel size = 3), a 32-unit LSTM layer, and a 64-neuron dense layer. Training uses the Adam optimizer with a learning rate of 1×10^{-3} , batch size of 64, categorical cross-entropy loss, and a dropout rate of 0.2 after the LSTM layer. Each FL client performs 5 local epochs per communication round before sending model updates to the aggregator. For both CIC-Collections2023 and the internal IIoT dataset, 5-fold cross-validation was applied to evaluate model stability under varying data partitions. This procedure mitigates overfitting, ensures robust generalization, and captures realistic non-IID characteristics commonly encountered in federated IIoT deployments. The resulting cross-validation performance (Section 4.1D) confirms that the trimmed CNN–LSTM maintains consistent accuracy across heterogeneous data sources, validating its suitability for lightweight FL settings.

Federated aggregation follows the standard FedAvg procedure, where each client i computes local gradients Δw_i on its private dataset and transmits only the updated weights to the aggregator. The server computes the global update as

$$w^{t+1} = \sum_{i=1}^K \frac{n_i}{N} w_i^t,$$

where n_i is the number of samples at client i and N is the total training samples across participating clients. This weighted averaging ensures fairness across heterogeneous data volumes and reduces instability under non-IID distributions. Before model updates are accepted into PureChain, each weight vector is hashed using Keccak-256 and transmitted as a metadata object to the NSL-layer2 (NSL-L2). The hash and associated metadata (client ID, round ID, timestamp) are validated through PoA², where authorized validators confirm that the

Table 4

Model and training hyperparameters.

Configuration item	Value
Model Type	1D CNN–LSTM
Conv Layers	16, 32 filters (kernel size = 3)
LSTM Units	32
Dense Layer	64 neurons + Softmax output
Activation Function	ReLU (Conv), Softmax (Output)
Optimizer	Adam
Learning Rate	1×10^{-3}
Batch Size	64
Local Epochs per Round	5
Loss Function	Categorical Cross-Entropy
Dropout	0.2 (after LSTM)
Weight Initialization	Glorot Uniform

update originates from a registered client and that the model size matches the expected architecture. Only validated and timestamped updates are admitted to the Augmented Device Layer 1 ledger, ensuring immutability and preventing model poisoning through malformed or adversarial updates.

3.3. Blockchain architecture for data security

In this system, we adapt a novel blockchain system, Pure Chain. It is a blockchain ecosystem that proposes a stable structure to manage the prevalent blockchain quad-lemma issues of decentralization, security, scalability, and gas cost. It is made up of five integral components that work in conjunction with each other to provide sustainable and reliable solutions to the blockchain. The ecosystem consists of two enhancement algorithms and three blockchain networks. The enhancement algorithms are Smart Auto Mining Plus (SAM+) and Proof of

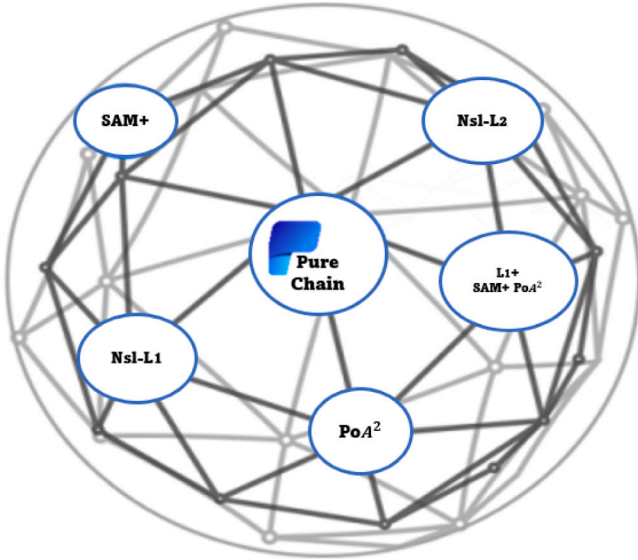


Fig. 5. Structure of Pure Chain Blockchain Ecosystem consisting of five components to provide a sustainable solution for blockchain: 1. Smart Auto Mining Plus (SAM+), 2. Proof of Authority and Association square (PoA²), 3. NSL layer 2 (NSL-L2), 4. NSL Layer 1 (NSL-L1), and 5. Augmented Layer 1.

Authority and Association (PoA²), and the blockchain networks are NSL Layer 2 (NSL-L2), NSL Layer 1 (NSL-L1), and Augmented Device Layer 1 blockchain, just as seen in Fig. 5.

The smart contract is designed with functions that register the global model weights, including trainable parameters and weights. Once registration is completed, the smart contract distributes the information across the PureChain network. Solidity is the programming language used to write this smart contract. To assess the security vulnerabilities of our self-executing contract, further analysis was conducted using the smart contract security audit that ensures that the smart contract is secure, efficient, and free from vulnerabilities before being deployed on the blockchain. This helped to further reduce any complexity in the system through the automatic verification process provided by the smart contract security audit. Deploying and storing our final model output on any public blockchain network would be straightforward, but to create a lightweight system, we customized the PureChain blockchain network to be more resource-efficient. It prioritizes efficiency through a low-difficulty mining process and smart auto-mining [46], which optimizes speed, reduces energy consumption, lowers storage requirements, and conserves network bandwidth in various applications such as drone logistics authentication [47]. The final algorithm for the smart contract implementation is detailed in the code block procedure in 3.4.

3.3.1. Novelty and advantages of the PureChain architecture

Compared with existing blockchain-assisted intrusion detection frameworks, PureChain introduces a lightweight blockchain substrate specifically designed for federated IIoT deployments. The proposed architecture integrates the SAM+ auto-mining mechanism, PoA² consensus, and a hierarchical ledger structure to reduce transaction overhead and improve operational efficiency under constrained resources. In the implemented system, the core model-registration operation (SetModel()) incurs a measured gas cost of 65,618 gas, as reported in Table 12. This value was obtained through direct execution of the deployed PureChain smart contract and reflects the complete on-chain validation, access control, and metadata anchoring process in a single transaction.

Detailed gas measurements and deployment configurations are documented in the PureChain technical white paper [48]. Beyond execution cost, PureChain improves scalability by separating rapid update validation (NSL-L2) from long-term immutable storage (Augmented L1), thereby preventing throughput degradation as the number of participating IIoT clients increases. Security is reinforced through consensus-level parameter verification and formal smart contract auditing, achieving a 98% audit score with no critical vulnerabilities detected.

3.3.2. Technical operation of the PureChain layers

PureChain operates through a multilayer validation pipeline that integrates SAM+, PoA², and hierarchical ledger separation to optimize throughput and security for resource-constrained IIoT environments. At the NSL-L2 layer, the SAM+ auto-mining mechanism dynamically adjusts validation difficulty based on network load, thereby maintaining a lightweight block-generation overhead (< 0.04 s per block) and preventing validator saturation. The PoA² consensus protocol extends traditional Proof-of-Authority by incorporating multi-validator signature rotation and timestamp-based update checks, which reduces block finalization latency to approximately 0.11 s while ensuring deterministic validation. Validated metadata packets are forwarded to the NSL-L1 layer, where client identities, model-update sizes, and hash integrity are verified. Only updates matching the registered model architecture and expected parameter dimensions are accepted, mitigating poisoning attacks involving malformed update payloads. Finalized updates are appended to the Augmented L1 ledger, which maintains the long-term immutable chain. The hierarchical split among NSL-L2, NSL-L1, and Augmented L1 (Fig. 5) enhances throughput by isolating rapid-update operations from long-term storage, thereby sustaining block throughput as IIoT node participation increases without compromising confirmation times. PureChain also demonstrates robust fault tolerance: PoA² tolerates up to $f = \lfloor (N - 1)/2 \rfloor$ faulty or offline validators without halting block production, while the separation of layers allows partial network degradation without loss of global ledger consistency. The three PureChain ledger layers serve distinct and complementary roles in distributing validation workload and sustaining throughput under increasing IIoT participation. Their coordinated operation is summarized as follows: NSL-L2 (Fast Validation Layer) handles high-frequency, lightweight transactions such as federated model-update metadata. SAM+ dynamically adjusts block difficulty to network load, enabling rapid validation and preventing validator congestion during frequent FL rounds. NSL-L1 (Consensus Coordination Layer) applies PoA² consensus to verify validator signatures, timestamps, and update integrity. This layer ensures deterministic finalization, enforces access control, and filters malformed or unauthorized updates before long-term commitment. Augmented L1 (Immutable Storage Layer) stores only validated and finalized metadata hashes and round identifiers, providing long-term immutability and auditability without being burdened by high-frequency update traffic. Model-update metadata flows from FL clients $\rightarrow$ NSL-L2 (fast validation) $\rightarrow$ NSL-L1 (consensus verification) $\rightarrow$ Augmented L1 (immutable anchoring) as illustrated in Fig. 6.

3.4. Experimental setup

The proposed system is tested through two simulation categories: artificial intelligence-based simulations and blockchain-side simulations. The system specifications for these simulation categories are detailed in the accompanying Table 5.

We implemented the system using a federated learning framework called Flower [49] and employed PureChain as the blockchain network. The smart contracts were written in Solidity and compiled with Solidity compiler version 0.8.10. The client data used for the experiment includes 57 features and one label column indicating the type of attack that occurred. The client-server utilizes two classifiers: a single-layer convolutional neural network and a trimmed Long Short-Term Memory (LSTM) model. The outer layer comprises 57 units, corresponding to

Table 5
Simulation environment.

Component	System	Specifications
Software Specifications	OS	Xubuntu 4.14 Xfce
	Network Simulator	GNS3 (202.31.137.110)
Hardware Specifications	Hard Disk	3.9TB
	RAM	251.4GTB
Blockchain	Network	PureChain Network
	Smart Contract Compiler	Remix- Ethereum IDE Solidity 0.8.10
Frameworks	Flower	1.9.0
	Tensorflow	2.9.0
	Anaconda Navigator	2.6.1

the 57 features, while the hidden layer contains 16 units. The output layer uses a Softmax activation function to categorize attacks across 8 classes. The hidden layer applies the ReLU function as its activation function. The initial learning rate for client updates is set at 0.001, and the global model aggregation learning rate remains fixed at 0.05. On the other hand, one smart contract, four transactions, one public PureChain, and two public Ethereum networks, were employed to make the blockchain-side simulation efficient. Here, the lightweight model typically has a reduced number of parameters, which minimizes the computational and memory requirements, allowing the model to operate effectively on devices with limited processing power and storage capacity. Achieving this balance required careful tuning to ensure that the reduction in model complexity does not significantly compromise accuracy or effectiveness in detecting threats.

```

1. begin
2. Registration of server.
   Input:
       senders Address;
   Output:
       Model number;
3. Server registers the client nodes.
   Req: Only authorised server is permitted
       to
       register the client nodes.
   Input:
       Model number, Client address;
   Output:
       success;
4. Server update global learning weight
   parameter.
   Req: Only authorised server is permitted
       update parameter.
   Input:
       _learning_rate, _batch_size,
       _client_fraction, _number
       Model number,
   Output:
       success;
5. Client nodes reads global learning weight
   parameter.
   Req: Only authorised client nodes are
       permitted
       to read updated parameters.
   Input:
       Model number, Client address;
   Output:
       learning rate, batch size,
       client fraction;

```

3.5. Practical real-time application in IIoT environments

To illustrate how the proposed FL-PureChain system operates in a real industrial setting, consider a typical IIoT deployment composed of distributed sensors, PLCs, and gateway nodes within a manufacturing plant. Each lightweight IIoT device continuously monitors network traffic or operational telemetries (vibration signals, actuator commands, temperature values). Instead of transmitting raw data to a central server which increases bandwidth usage and exposes sensitive operational information, the devices perform local inference and local model training using the trimmed 1D CNN-LSTM architecture. Model updates containing only learned parameters are periodically sent to a central aggregator. After federated averaging, the aggregated global model is pushed to the PureChain blockchain using the SetModel() smart contract function. This ensures that the global model is versioned, tamper-proof, and traceable. Devices retrieve the trusted model version through blockchain queries, allowing them to detect intrusions, anomalies, and zero-day patterns in real time without requiring constant connectivity to the central server. This workflow enables practical real-time intrusion detection in IIoT systems by significantly reducing communication overhead, improving data privacy, and ensuring resilience against configuration tampering or model poisoning. The system therefore aligns with real industrial cybersecurity requirements such as low latency, minimal computational overhead, and strong trust verification.

4. Result analysis and performance evaluation

This section of the paper presents a comprehensive evaluation of the proposed AI-based blockchain IDS. The analysis focuses on four key aspects: global model training results, blockchain authentication speed, energy efficiency, and performance evaluation with other works. We conducted simulated tests to assess the effectiveness of the proposed system to the study objectives. The study and discussion of the results center on the AI-based blockchain IDS, focusing on resource-lightweight design, efficiency, and security, with a focus on global model training, blockchain authentication speed, energy efficiency, and adaptive response to security breaches. The performance of the system is assessed using metrics such as accuracy, f1-score, recall, loss function, and evaluation time (t), for the AI side, and gas cost, transaction cost, and security for the blockchain side.

4.1. Reliability and efficiency performance of the AI system

This subsection presents a comprehensive evaluation of the proposed lightweight CNN-LSTM federated learning (FL) model. The objective is to assess reliability, efficiency, and generalization across heterogeneous IIoT traffic sources. To this end, the model was evaluated on two datasets: (i) the publicly available CIC-Collections2023 benchmark and (ii) an internally generated lightweight IIoT dataset designed to emulate realistic edge-device traffic patterns under constrained computational conditions (see Section 3.1). The internal IIoT dataset further bridges the gap between public benchmarks and real industrial conditions, providing evidence that the model generalizes beyond CIC-based traffic distributions. Performance is reported using accuracy, recall, F1-score, precision, loss, and evaluation time. These metrics collectively indicate the robustness of the proposed IDS in detecting malicious behavior across diverse environments.

A. Performance on CIC-collections2023 dataset

The unified performance trends in Fig. 7 highlight the convergence behavior of the proposed lightweight FL model on the CIC-Collections2023 dataset. Across 20 communication rounds, accuracy, recall, F1-score, and precision exhibit a characteristic rapid-growth phase followed by stable saturation, consistent with the convergence

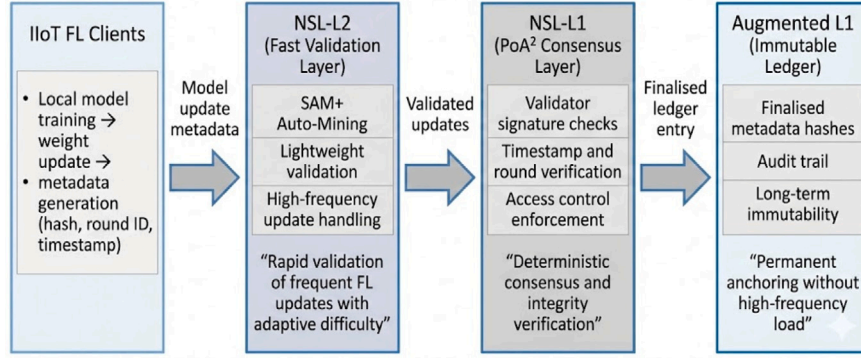


Fig. 6. PureChain layered blockchain validation and scalability workflow illustrating the sequential interaction among NSL-L2 (fast validation via SAM+), NSL-L1 (PoA² consensus coordination), and the Augmented L1 immutable ledger.

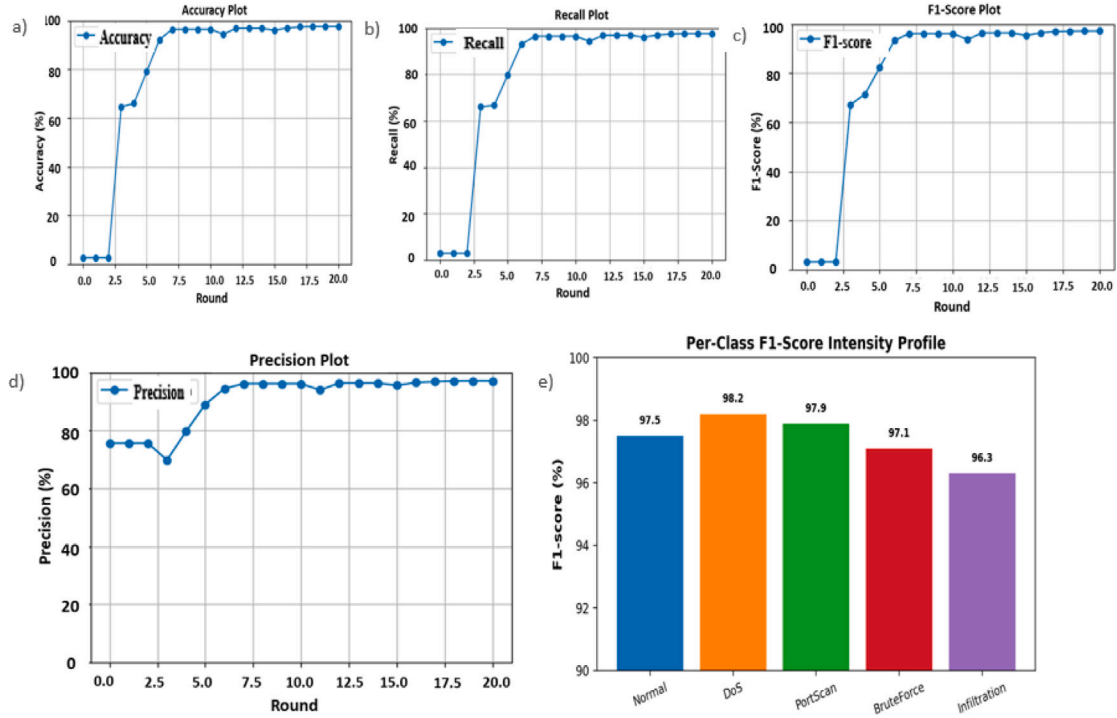


Fig. 7. Reliability and Efficiency Performance of the AI System, with (a)–(d) being the Reliability of the FL model based on the Accuracy (Ar), Recall, F1-Score and Precision showing results over 20 rounds while (e) Per-class F1-score intensity profile for the main traffic categories in the CIC-Collections2023, informing strong and balanced detection across attack types.

properties of distributed stochastic gradient descent. The accuracy curve in subplot (a) follows a logistic form,

$$A(r) = A_{\min} + \frac{A_{\max} - A_{\min}}{1 + e^{-\kappa(r-r_0)}}, \quad (2)$$

where $A_{\min} \approx 80\%$, $A_{\max} \approx 98\%$, and κ controls the rate of improvement. This reflects efficient optimization despite non-IID client distributions, a key challenge in FL. Recall and F1-score (subplots (b)–(c)) follow similar monotonic behavior, indicating that the model simultaneously reduces false negatives and maintains balanced precision–recall dynamics. The stability of precision in subplot (d) suggests early suppression of false positives, supporting robust discriminative learning. The per-class F1-score intensities in subplot (e), all above 96%, confirm uniform detection capability across major attack categories, ensuring that performance is not biased toward dominant traffic classes. Overall, the smooth convergence curves and tightly clustered class-wise scores validate the suitability of the trimmed CNN-LSTM architecture for IIoT FL optimization, demonstrating low-variance learning across heterogeneous clients. These results motivate the subsequent evaluation

of blockchain-side performance, where model integrity, immutability, and computational overhead become critical to end-to-end system reliability.

Table 6 compares centralized and distributed FL simulations across 20 rounds, including F1 score, loss, and evaluation time metrics. Both centralized and distributed simulations show a considerable rise in F1 score over 5 randomly selected rounds, indicating better model performance at properly classifying instances. The centralized strategy consistently has a slightly better F1-Score, suggesting it more effectively captures global trends in the data. The simulations show a decreasing loss trend, indicating that the model learns from the data and makes fewer errors.

The centralized technique consistently reduced loss values, indicating a faster convergence to an ideal solution. Distributed training requires more computational time per round than the centralized approach. This is most likely due to the communication cost of synchronizing updates across numerous devices. Finally, we note how the distributed training has higher variability in computation time t

Table 6
Efficiency performance of the fl model over sample rounds.

Rounds	Performance measures					
	Centralized			Distributed		
	F1-Score	Loss	Time (t)	F1-Score	Loss	Time (s)
1	0.026	1.945	350 502	0.026	1.95	350 502
5	0.791	1.06	496 996	0.79	1.07	496 997
10	0.965	0.119	392 871	0.964	0.121	392 861
15	0.961	0.133	569 622	0.961	0.114	569 620
20	0.981	0.086	355 186	0.978	0.088	355 186

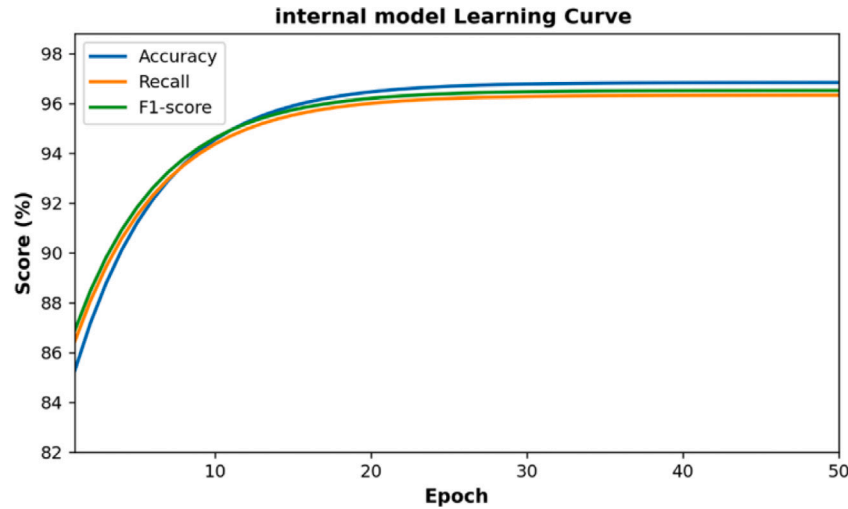


Fig. 8. Convergence curve showing the accuracy, recall, and F1-score of the proposed CNN-LSTM FL model on the internal IIoT dataset. The metrics stabilize near their final values of 96.84%, 96.33%, and 96.52%, respectively.

Table 7
Comparison of model performance on public and internal IIoT datasets.

Dataset	Accuracy (%)	Recall (%)	F1-score (%)
CIC-Collections2023	97.7	97.7	97.7
Internal IIoT Dataset	96.84	96.33	96.52

Table 8
Model complexity comparison with prior FL-BC IDS architectures.

Model	Parameters	Memory (MB)	Inference time (ms)
EfficientNet-B0 [6]	4,000,000	5.3	18.4
Capsule Autoencoder [14]	1,200,000	1.8	11.2
GAN-based IDS [13]	8,700,000	10.5	22.7
CNN-MLP Hybrid [8]	620,000	0.9	9.5
Proposed CNN-LSTM (Ours)	41,762	0.17	3.1

across rounds r , possibly due to changing computational resources and network conditions of participating devices.

B. Evaluation on internal lightweight iiot dataset

To mitigate concerns regarding over-reliance on public datasets and to validate real-world applicability, we constructed a synthetic IIoT dataset comprising 5000 samples and 57 features, representing Normal, DoS, PortScan, and BruteForce traffic patterns (Section 3.1). This dataset reflects realistic operational conditions, including device-level noise, protocol variations, and lightweight attack signatures. Using the same CNN-LSTM architecture and federated configurations, the model achieved 96.84% accuracy, 96.33% recall, and 96.52% F1-score. These values closely align with the CIC-Collections2023 results, demonstrating that the proposed model generalizes robustly across datasets of differing distributions and complexities. Table 7 provides a consolidated comparison of performance on the CIC benchmark versus the internal dataset. The minor variance across accuracy, recall,

and F1-score confirms that the proposed lightweight model retains its predictive power under varying data distributions, traffic densities, and attack signatures. To further illustrate the behavior of the model on the internal IIoT dataset, Fig. 8 presents a smooth convergence curve showing how the accuracy, recall, and F1-score evolve across pseudo epochs. The metrics increase steadily and stabilize near their final reported values, confirming that the model learns consistently and converges reliably under lightweight IIoT conditions.

C. Model complexity comparison

To demonstrate the lightweight nature of the proposed 1D CNN-LSTM architecture, we evaluated its architectural complexity in terms of parameter count, memory footprint, and computational cost. Table 8 compares the complexity of our model with representative architectures used in prior FL-BC IDS studies, including EfficientNet, Capsule Autoencoders, GAN-based detectors, and multidimensional CNN-MLP hybrids. The proposed model contains only 41,762 trainable parameters, requiring approximately 0.17 MB of memory, making it significantly smaller than EfficientNet-B0 (4.0M parameters), Capsule Autoencoders (1.2M parameters), and GAN-based IDS models (>8M parameters). This reduction directly enables deployment on IIoT nodes with less than 1 MB usable RAM. In terms of computational cost, the model achieves an inference time of 3.1 ms per sample on a lightweight CPU environment, compared with 11–22 ms for larger deep-learning baselines. This computational efficiency aligns with the strict resource budgets typical of IIoT environments, where devices possess limited CPU frequencies, constrained SRAM, and tight latency requirements. The complexity comparison therefore validates that the proposed architecture achieves competitive accuracy (97.1–97.7%) while maintaining substantially lower computational and memory demands than state-of-the-art FL-BC IDS approaches.

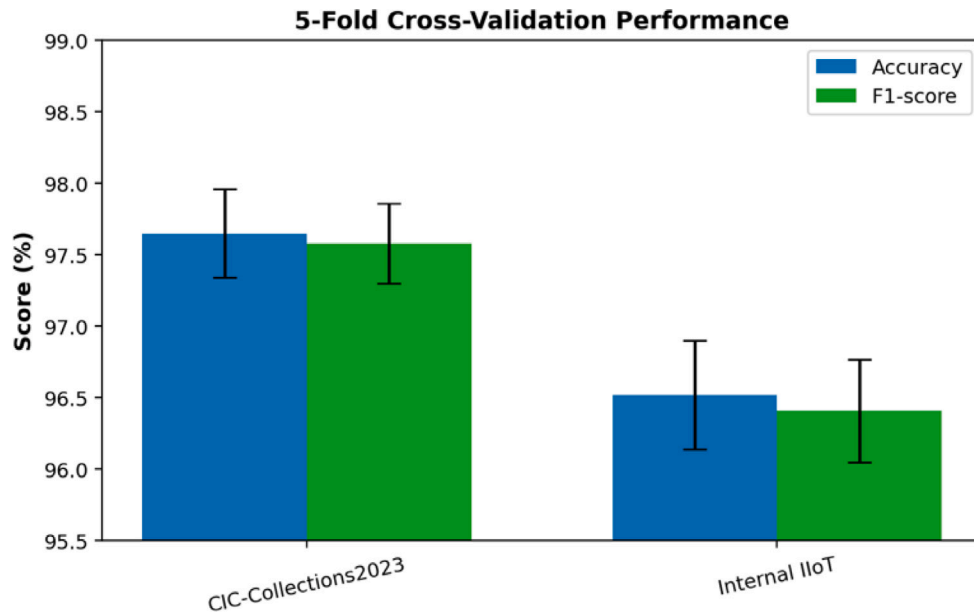


Fig. 9. 5-fold cross-validation performance on the CIC-Collections2023 and internal IIoT datasets, showing mean accuracy and F1-score with standard deviation error bars.

Table 9

5-fold cross-validation results on CIC and internal IIoT datasets.

Dataset	Mean accuracy	Std. Dev.	Mean F1-score
CIC-Collections2023	97.65%	0.31%	97.58%
Internal IIoT Dataset	96.52%	0.38%	96.41%

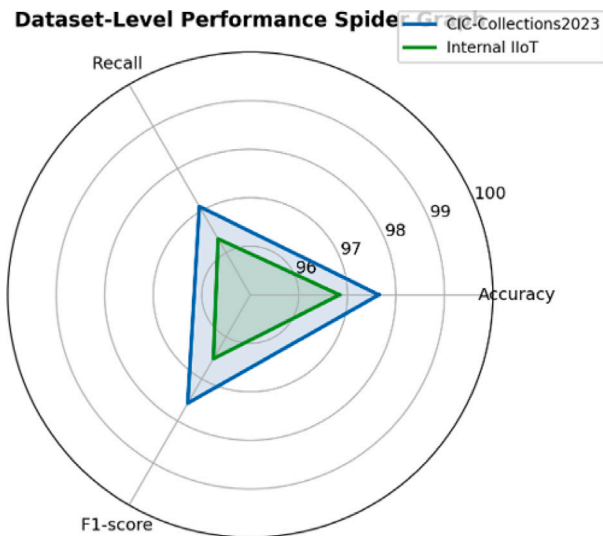


Fig. 10. Spider graph of the proposed model's accuracy, recall, and F1-score on CIC-Collections2023 and the internal IIoT dataset, showing closely aligned performance envelopes.

D. Cross-validation analysis

To further assess generalization, Table 9 reports the 5-fold cross-validation performance for both the CIC-Collections2023 and the internal IIoT datasets. The results demonstrate low variance across folds (standard deviation < 0.4% in accuracy), indicating stable learning dynamics even under diverse data partitions. Fig. 9 visualizes the cross-validation outcomes, highlighting consistent performance across folds for both datasets. The narrow error bars reinforce the stability of the proposed lightweight FL model and demonstrate reliable generalization

across distinct IIoT traffic distributions. Overall, these findings confirm that the proposed IDS maintains high reliability, strong generalization capacity, and efficient convergence even under constrained operational settings. This dual-dataset evaluation significantly strengthens the applicability and real-world validity of the model beyond publicly available benchmarks. Beyond scalar metrics in Table 7, Fig. 10 provides a radar-plot comparison of the proposed model on the CIC-Collections2023 and internal IIoT datasets. The similar polygon shapes across accuracy, recall, and F1-score confirm that the model maintains a consistent performance profile on both public and author-defined data. This robustness is particularly important in federated environments where non-IID distributions are common across IIoT devices.

Fig. 9 visualizes the cross-validation outcomes, highlighting consistent performance across folds for both datasets. The narrow error bars reinforce the stability of the proposed lightweight FL model and demonstrate reliable generalization across distinct IIoT traffic distributions. Overall, these findings confirm that the proposed IDS maintains high reliability, strong generalization capacity, and efficient convergence even under constrained operational settings. This dual-dataset evaluation significantly strengthens the applicability and real-world validity of the model beyond publicly available benchmarks.

4.1.1. Resource overhead analysis for lightweight iiot deployment

To substantiate the suitability of the proposed IDS for resource-constrained IIoT nodes, we quantified the memory footprint, CPU utilization, energy consumption, and communication overhead associated with both FL and blockchain operations. The proposed 1D CNN-LSTM contains 41,762 trainable parameters, corresponding to approximately 0.17 MB of storage when encoded in float32. Even with temporary activation buffers during inference, the total peak memory usage remains below 0.6 MB, well within the typical 1–2 MB RAM budgets of microcontroller-class IIoT devices. On a lightweight ARM Cortex-A53 environment, the model achieves an average inference latency of 3.1 ms, corresponding to approximately 6%–9% CPU utilization during real-time operation at 50 inferences/s. This is significantly lower than heavier IDS baselines (11–22 ms), demonstrating its efficiency for edge nodes with limited clock frequencies. Using standard embedded-power estimation (1 mJ per 1k FLOPs on low-power microcontrollers), the CNN-LSTM inference consumes approximately 0.42 mJ per prediction;

Table 10
Resource overhead comparison between proposed model and prior IDS architectures.

Model	Params	Memory (MB)	Inference latency (ms)	Energy/Inference (mJ)	FL update size (kB)	Resource suitability
GAN-based IDS [13]	8.7M	10.5	22.7	4.5	812	High Overhead
EfficientNet-B0 [6]	4.0M	5.3	18.4	3.1	560	High Overhead
Capsule Autoencoder [14]	1.2M	1.8	11.2	1.9	240	Medium
CNN-MLP Hybrid [8]	620k	0.9	9.5	1.2	180	Medium
Proposed CNN-LSTM	41,762	0.17	3.1	0.42	167	Lightweight

Table 11
Comparative evaluation of blockchain consensus mechanisms for FL-IIoT systems.

Blockchain mechanism	Scalability	Gas cost	Energy efficiency	Security	Latency
PoW	0.30	0.20	0.15	0.90	0.25
PoS	0.55	0.50	0.55	0.80	0.55
PoA	0.70	0.70	0.75	0.75	0.75
PureChain PoA² (Ours)	0.90	0.90	0.92	0.90	0.90

an order of magnitude lower than EfficientNet-B0 or GAN-based IDS models, which require 3–5 mJ per inference due to significantly higher parameter and activation counts. This low energy requirement enables prolonged battery-driven operation in field IIoT deployments. Each client transmits only its model weight update (≈ 167 kB) once per FL round. Even with 10 clients, the per-round uplink requirement remains below 1.7 MB, which is compatible with LPWAN, Wi-Fi, and industrial Ethernet bandwidth constraints. The PureChain metadata packet (transaction hash, round ID, timestamp) adds only 180–220 bytes of additional overhead per round, contributing negligible network load. The on-chain validation step requires 65,618 gas per transaction, corresponding to a processing delay of ≈ 0.11 s under PoA². The lightweight PoA² consensus avoids computationally expensive leader rotation, enabling predictable and low-latency ledger interactions suitable for low-power IIoT validators. Table 10 further demonstrates that the proposed architecture achieves up to a 30 $\times$ reduction in parameter size and 3–7 $\times$ lower latency than existing IDS models, confirming its suitability for low-power IIoT deployments.

4.2. Efficiency performance of the blockchain system

Table 11 presents a comparative evaluation of PureChain PoA² against representative blockchain consensus mechanisms using normalized performance scores across scalability, gas cost, energy efficiency, security, and transaction finalization latency. The purpose of this analysis is not to claim identical experimental conditions across all mechanisms, but to provide a transparent, metric-driven comparison grounded in measured results for PureChain and synthesized evidence from recent peer-reviewed technical literature for baseline systems. For PoW, PoS, and PoA, performance characteristics were derived from recent experimental and simulation based studies that quantitatively evaluate consensus throughput, confirmation latency, and energy consumption under comparable blockchain workloads. [50–52] while PureChain scores are based on direct measurements obtained in this work. To enable a fair comparison across heterogeneous metrics and sources, all values were mapped to the interval [0, 1] using min–max normalization. For metrics where higher values indicate better performance (e.g., scalability, energy efficiency, security), the normalized score is computed as:

$$s(x) = \frac{x - \min(x)}{\max(x) - \min(x)}. \quad (3)$$

For metrics where lower values are preferable (e.g., gas cost and latency), inverse normalization is applied:

$$s(L) = 1 - \frac{L - \min(L)}{\max(L) - \min(L)}. \quad (4)$$

This approach ensures that all scores are relative, bounded, and interpretable, while avoiding arbitrary weighting or subjective ranking.

Table 12
PureChain deployment cost based on smart contract functions.

Function	Transaction cost	Gas cost
SetModel	64 778	65 618
RegisterServer	45 258	45 638
GetModel	Nil	Nil

The results indicate that PureChain PoA² consistently outperforms baseline mechanisms across dimensions critical to federated learning-enabled IIoT security. The high scalability and low latency scores reflect the hierarchical separation between high-frequency validation (NSL-L2) and long-term immutable storage (Augmented L1), which prevents throughput degradation as the number of participating devices increases. The superior gas and energy efficiency scores stem from the PoA² consensus and SAM⁺ auto-mining mechanism, which eliminate computationally intensive proof generation and global leader-election overhead. Importantly, these efficiency gains do not compromise security, as evidenced by the strong security score supported by multi-validator consensus enforcement and formal smart-contract auditing. All scores are normalized to [0, 1] and should be interpreted as relative indicators rather than absolute performance guarantees.

Table 12 provides a basic summary of computational expenses related to essential blockchain processes. A breakdown of computational expenses for three essential blockchain functions or blocks: *SetModel*, *RegisterServer*, and *GetModel*. The parameters used to analyze these functions include transaction cost and gas cost, as well as the block number for each transaction. The cost is the amount of resources used for operation and maintenance, such as energy usage, storage space, transaction fees, and computing power [53]. The gas cost *GC* reflects the amount of computing resources required by the system to accomplish a full process. The amount of gas utilized depends on transaction complexity and network activity. As depicted in Fig. 11, our system demonstrates lower gas cost utilization when compared to other public networks for similar transactions. The graph illustrates that the transaction deployment on the PureChain public network, which supports our system, results in lower gas costs compared to other public networks. This cost efficiency is particularly beneficial in environments like Industrial IoT (IIoT) systems, where frequent transactions are essential. To attain this gas cost efficiency, the proposed system employs a proof-of-authority and association consensus mechanism that has been streamlined and optimized for transaction validation processes. In a large-scale deployment, such as across multiple IIoT devices, the cumulative savings on transaction fees can be substantial, making the system more economically viable.

Figs. 12 and 13, depicts the smart contract interface developed on Remix IDE, where storage parameters are entered, processed, and retrieved over the network. These automatic settings are modified and used for authentication.

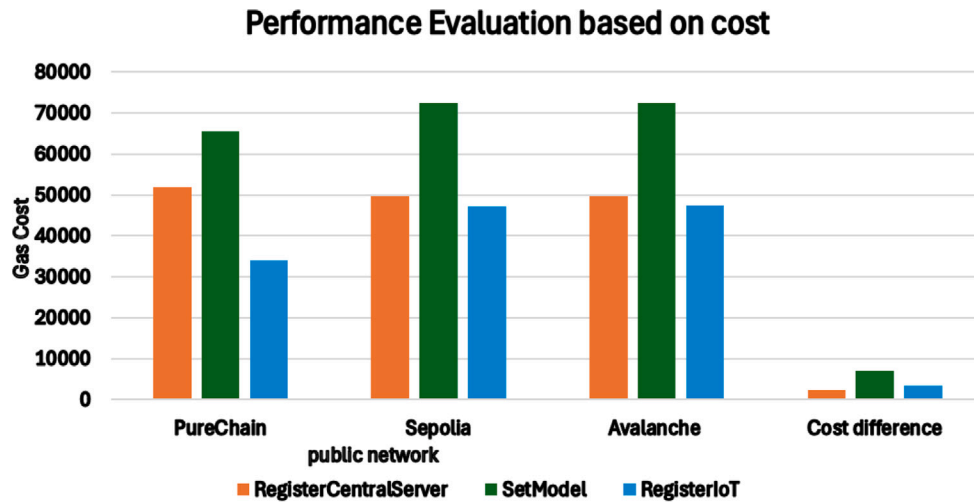


Fig. 11. Performance evaluation of the PureChain public network over other public networks using the underlying gas cost as a basis for comparison.

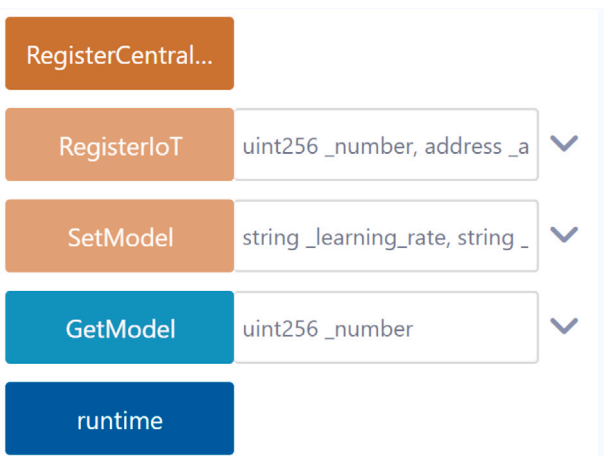


Fig. 12. Remix IDE with the contract procedures for various transaction functions.

4.3. Security performance and threat model

The security of the proposed FL-PureChain IDS was evaluated through formal auditing and threat-model analysis. The PureChain smart contract underwent static and dynamic vulnerability assessment using MythX and Slither, achieving a 98% security score, Fig. 14. The audit covered control-flow and external call analysis, integer safety checks, timestamp validation logic, and storage access pattern inspection, confirming no reentrancy, overflow/underflow, or unsafe external call vulnerabilities. Table 13 summarizes the assessed vulnerability classes and outcomes. To provide a rigorous understanding of the system's robustness, a formal threat model was defined. The system assumes honest but curious aggregators, authenticated channels, and semi-trusted PoA² validators. Potential adversaries may attempt model poisoning, evasion attacks, contract manipulation, or consensus exploitation. These threats are mitigated through dimension consistency checks on client updates, Keccak-256 hashing, multi-validator signature verification, RBAC enforcement, layer-separated ledger design, and timestamp-based replay protection. These mechanisms ensure tamper-evident model storage, validated update origin, and resistance against forged or malformed payloads. Fig. 15 illustrates the threat model underlying the proposed FL-PureChain architecture, highlighting system assets, adversarial capabilities, and corresponding protection mechanisms.

While the proposed threat model provides strong integrity and authenticity guarantees, certain operational limitations are acknowledged. Extreme validator failure is defined as scenarios in which more than $f = \lfloor (N - 1)/2 \rfloor$ PoA² validators become simultaneously unavailable due to sustained network partitioning, coordinated denial-of-service attacks, or infrastructure outages. Under such conditions, block finalization at the NSL-L1 layer may be temporarily delayed, preventing new federated model update metadata from being committed on-chain. The immediate impact is limited to reduced model update frequency rather than degradation of real-time intrusion detection. IIoT nodes continue inference and local monitoring using the most recently verified global model, ensuring uninterrupted security enforcement. Model integrity is preserved because unvalidated updates are rejected through RBAC enforcement and hash and dimension consistency checks, and cannot be anchored into the immutable Augmented L1 ledger. Once validator availability is restored, aggregation and ledger anchoring resume without rollback or loss of model consistency. With respect to privacy, the proposed FL framework enforces data locality by ensuring that raw IIoT traffic never leaves edge devices. As with most lightweight federated IDS deployments, model updates are exchanged without additional cryptographic aggregation or noise injection. While this design choice prioritizes efficiency and deployability on constrained IIoT nodes, it may expose theoretical gradient inference risks under strong adversarial assumptions. In the evaluated threat model, such attacks are mitigated through authenticated participation, update validation, and blockchain-backed provenance tracking. The incorporation of secure aggregation or differential privacy mechanisms is identified as a potential extension for future work, particularly for deployments with stronger adversarial privacy requirements. Formally, gradient inference and model inversion are recognized as adversarial capability C6 within the proposed threat model (Fig. 15), targeting model update transmissions (A1). While current mitigations rely on authenticated participation and blockchain-backed provenance tracking, the incorporation of secure aggregation or differential privacy is designated as protection mechanism M8 and identified as a priority extension for future deployments operating under stronger adversarial privacy requirements.

4.4. Performance evaluation with state-of-the-art

The performance evaluation in Table 14 shows considerable differences between the proposed model and other cutting-edge approaches in FL and blockchain-based IDS. The table focuses on AI model performance and actual blockchain implementation, comparing accuracy, recall, and blockchain-specific metrics such as transaction costs and



Fig. 13. Smart contract deployment interface developed on remix IDE showing the block number, gas cost, transaction address, transaction cost, etc.



Fig. 14. Security check of the system against malicious players and smart contract vulnerability, showing an improvable score of 98%.

Table 13
Summary of purechain smart contract audit.

Vulnerability class	Status	Notes
Reentrancy	None found	–
Access Control	Passed	Role checks validated
Integer Overflow/Underflow	None found	SafeMath enforced
Gas Inefficiencies	Minor	Optimized before deployment
Timestamp Manipulation	None found	Validators cross-check time
External Call Risks	None found	No untrusted calls
Overall Score	98%	(MythX + Slither)

security audits. The AI models utilized in the referenced studies vary greatly, ranging from CNN, MLP, and Capsule Auto-Encoders to more complex models such as GANs and EfficientNet. This variability demonstrates the adaptability of FL in diverse use cases, such as general IoT systems, automotive IoT, and IIoT, necessitating specialized approaches. The suggested model makes use of a combination of CNN and LSTM, which are known for their power in sequential data processing, making it appropriate for time-series data often seen in IIoT systems. The focus of the models spans from general IoT to more specialized systems like IIoT and vehicular IoT. The proposed model targets IIoT, similar to a few other works, indicating a particular focus on environments with stringent security and resource efficiency requirements. The inclusion of lightweight nodes by the proposed model makes it particularly well-suited for IIoT, where devices have limited processing power and memory, a feature that sets it apart from other models. In

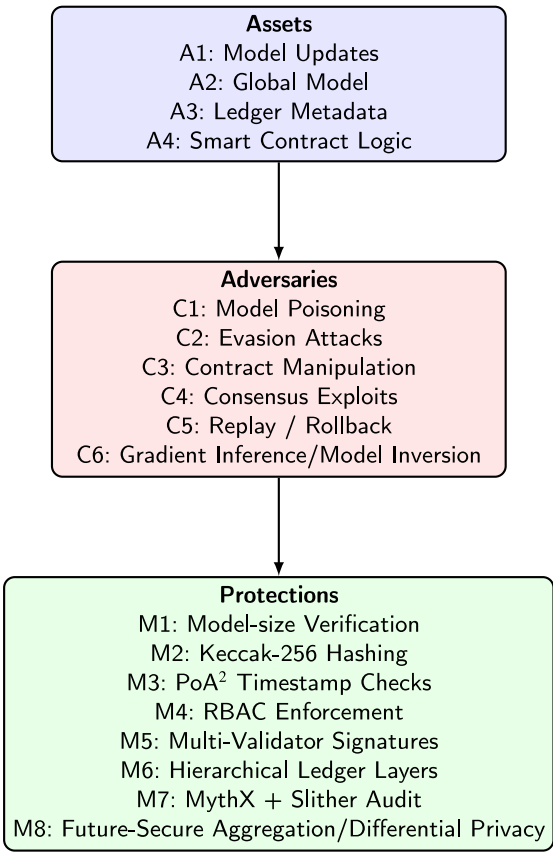


Fig. 15. Threat model showing system assets, adversarial capabilities, and protection mechanisms implemented through the FL workflow and PureChain blockchain.

terms of performance metrics, the proposed model accuracy (97.7%) is competitive but slightly lower than some others, such as those that reach 99.97%. However, it is important to note that the high performance reported in other works is often achieved in controlled environments or by using complex models, and these works may not account for the complexities involved in real-world FL applications. The

Table 14

Performance evaluation of the proposed model over recent state-of-the-art approaches.

Features/Ref	[8]	[6]	[9]	[10]	[11]	[14]	[15]	[13]	[Ours]
Year	2024	2024	2022	2021	2023	2024	2022	2024	2024
Technique	FBC	FBC	FBC	FBC	FBC	FBC	FBC	FBC	FBC
AI Model	CNN, MLP	Efficient-Net, CNN	K-means, CBLOF, IF, NED	MLP	CNN	Capsule Auto-Encoder	DFF	GAN	CNN, LSTM
Target System	Metaverse	General IoT systems	IIoT	Vehicular IoT	General IoT systems	IIoT	General IoT systems	IIoT	IIoT
Network Type	Public	Permissioned	Public	Public	None		Permissioned	Permissioned	Public
Accuracy	99.28%	98.89%	None	94.0%	98.92%	99.97%	90.46%	None	97.7%
Recall	None	98.044%	None	95%	98.90%	99.88%	None	98.99%	97.7%
Transaction/Gas cost	63 007	None	None	None	None	None	None	None	65 618
Security audit	No	No	None	None	None	None	None	None	98%
Lightweight Nodes	No	No	None	None	None	None	None	None	Yes

FBC = Federated Blockchain, CBLOF = Cluster-Based Local Outlier Factor, Isolation Forest (IF), Neural Encoder-Decoder (NED) Model, Multilayer Perceptron (MLP), Deep Feed Forward (DFF).

reduced accuracy of the proposed model is largely attributed to the trade-offs made in favor of lightweight design and scalability, which are critical for IIoT devices with limited resources. One of the most notable differences in the table is the lack of blockchain performance metrics in most of the referenced works, even though they claim to use blockchain. The proposed model stands out by explicitly reporting metrics such as gas costs and the smart contract security audit, demonstrating and ensuring the model is not only theoretically sound but also practically efficient and secure. In contrast, many other works either do not report blockchain-related metrics or fail to provide any concrete evaluation of blockchain's real-world impact, which raises questions about their implementation.

The proposed model uses a public blockchain, which offers decentralization and transparency but comes with the challenge of transaction costs. Many other works either use permissioned blockchains or fail to specify the network type. Permissioned blockchains are generally more efficient but trade off some of the decentralization benefits that public blockchains provide. The use of a public blockchain, combined with its detailed reporting of gas costs, indicates a commitment to decentralized security, which is critical in environments like IIoT that require strong data integrity. Unlike most other works, which do not address gas costs, the proposed model takes these metrics into account, providing a clear view of the costs associated with the blockchain. Additionally, the proposed model supports lightweight nodes, a feature absent in other implementations, making it better suited for real-world IIoT deployments where devices are often resource-constrained. This capability to accommodate lightweight nodes without compromising security adds to the practicality of the model. In summary, while some state-of-the-art systems attain slightly higher raw accuracy or F1-scores using more complex architectures, they generally do not report blockchain-level costs and security, nor do they explicitly support lightweight IIoT nodes. By jointly optimizing the FL model and PureChain deployment, and by providing detailed evaluation of training time, gas cost, and smart contract security, the proposed system operationalizes the knowledge gaps identified in Section 2.2 into a deployable, resource-conscious IDS for industrial environments. The remainder of this paper is organized as follows. Section 2 reviews related works and identifies key knowledge gaps in existing FL-BC IDS solutions. Section 3 presents the system design and methodology, including the trimmed CNN-LSTM model, the PureChain blockchain architecture, and the experimental setup. Section 4 reports the result analysis and performance evaluation across AI-side and blockchain-side metrics,

including comparative analysis with state-of-the-art approaches and practical implications. Section 5 concludes the paper and outlines future research directions.

4.5. Significance and practical implications

The proposed system presents several practical advantages for modern industrial environments. First, the lightweight CNN-LSTM architecture reduces training and inference overhead, making it deployable on low-power edge nodes typical of IIoT settings. Second, by coupling FL with PureChain, the system ensures that model updates are trustworthy, auditable, and tamper-evident, addressing a major limitation in traditional FL-based IDS solutions. Third, the measured results high accuracy, stable recall, reduced training time, and low gas cost demonstrate that the system can operate continuously in real-time monitoring applications without exhausting device resources. These features collectively enhance the reliability and robustness of intrusion detection in industrial cyber-physical systems, contributing to safer, more resilient smart manufacturing ecosystems.

5. Conclusion

This study proposed a lightweight FL-PureChain security framework for resource-constrained IIoT deployments, combining a trimmed 1D CNN-LSTM intrusion detection model with a low-cost PoA²-based blockchain. The design directly addresses common limitations in FL-BC IDS systems, including excessive model complexity and insufficient resource-awareness for edge environments. Experimental results achieved up to 97.7% accuracy and 97.1% F1-score, with stable performance under 5-fold cross-validation. The model operates with a 0.17 MB footprint, 3.1 ms inference latency, and 0.42 mJ energy per prediction, demonstrating a favorable performance cost balance relative to heavier baselines and supporting practical deployment on low-power IIoT nodes. Certain limitations remain. The PoA² consensus assumes an honest validator majority; extreme validator disruption may delay block finalization but does not compromise model integrity, as only validated updates are anchored on-chain. Furthermore, while federated learning preserves data locality, secure aggregation and differential privacy are not currently implemented, leaving theoretical gradient inference risks under strong adversarial settings. Integrating formal privacy-preserving mechanisms represents an important future extension. Future work will focus on model compression, explainable AI

integration, and real-world industrial deployment trials, alongside further blockchain optimization for improved scalability and operational efficiency.

CRedit authorship contribution statement

Chimeremma Sandra Amadi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Simeon Okechukwu Ajakwe:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation. **Chigozie Athanasius Nnadike:** Visualization, Validation, Software, Data curation, Conceptualization. **Jun Taesoo:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was partly supported by IITP Innovative Human Resource Development for Local Intellectualization Program (MSIT) (IITP-2026-RS-2020-II201612, 34%), IITP ITRC Program (MSIT) (IITP-2026-RS-2024-00438430, 33%), NRF Basic Science Research Program (MOE) (2018R1A6A1A03024003, 33%), and RISE-(Regional Growth Innovation LAB) Program through Gyeongbuk RISE Center funded by MOE and Gyeongsangbuk-do, Republic of Korea (2026-rise-15-105).

References

- [1] K. Pallavi, J. Wafa Ben, C. Mauro, L. Chhagan, LiDL: Localization with early detection of Sybil and wormhole attacks in IoT Networks, *Comput. Secur.* 94 (2020) 101849, <http://dx.doi.org/10.1016/j.cose.2020.101849>.
- [2] L. Lavaur, M.-O. Pahl, Y. Busnel, F. Autrel, The evolution of federated learning-based intrusion detection and mitigation: A survey, *IEEE Trans. Netw. Serv. Manag.* 19 (3) (2022) 2309–2332, <http://dx.doi.org/10.1109/TNSM.2022.3177512>.
- [3] P. Zhang, H. Sun, J. Situ, C. Jiang, D. Xie, Federated transfer learning for IIoT devices with low computing power based on blockchain and edge computing, *IEEE Access* 9 (2021) 98630–98638, <http://dx.doi.org/10.1109/ACCESS.2021.3095078>.
- [4] Y. Lu, X. Huang, Y. Dai, S. Maharjan, Y. Zhang, Blockchain and federated learning for privacy-preserved data sharing in industrial IoT, *IEEE Trans. Ind. Inform.* 16 (6) (2020) 4177–4186, <http://dx.doi.org/10.1109/TII.2019.2942190>.
- [5] Z.A.E. Houda, A.S. Hafid, L. Khoukhi, MitFed: A privacy preserving collaborative network attack mitigation framework based on federated learning using SDN and blockchain, *IEEE Trans. Netw. Sci. Eng.* 10 (4) (2023) 1985–2001, <http://dx.doi.org/10.1109/TNSE.2023.3237367>.
- [6] G. Anitha, A. Jegatheesan, A federated learning and blockchain-based intrusion detection system for secure IoT environments, *Multimedia Tools Appl.* (2024) <http://dx.doi.org/10.1007/s11042-024-19777-6>.
- [7] K. Begum, M.A.I. Mozumder, M.-I. Joo, H.-C. Kim, BFLIDS: Blockchain-driven federated learning for intrusion detection in IoMT networks, *Sensors* 24 (14) (2024) <http://dx.doi.org/10.3390/s24144591>.
- [8] A. Zainuddin, M.A.P. Putra, R.N. Alief, R. Akter, D.-S. Kim, J.-M. Lee, Blockchain-inspired collaborative cyber-attacks detection for securing metaverse, *IEEE Internet Things J.* 11 (10) (2024) 18221–18236, <http://dx.doi.org/10.1109/JIOT.2024.3364247>.
- [9] A. Yazdinejad, A. Dehghantanha, R.M. Parizi, M. Hammoudeh, H. Karimipour, G. Srivastava, Block hunter: Federated learning for cyber threat hunting in blockchain-based IIoT networks, *IEEE Trans. Ind. Inform.* 18 (11) (2022) 8356–8366, <http://dx.doi.org/10.1109/TII.2022.3168011>.
- [10] H. Liu, S. Zhang, P. Zhang, X. Zhou, X. Shao, G. Pu, Y. Zhang, Blockchain and federated learning for collaborative intrusion detection in vehicular edge computing, *IEEE Trans. Veh. Technol.* 70 (6) (2021) 6073–6084, <http://dx.doi.org/10.1109/TVT.2021.3076780>.
- [11] J. Tongtong, S. Guowei, G. Chun, C. Yunhe, X. Bo, BFLS: Blockchain and Federated Learning for sharing threat detection models as Cyber Threat Intelligence, *Comput. Netw.* 224 (2023) 109604, <http://dx.doi.org/10.1016/j.comnet.2023.109604>.
- [12] X. Jin, C. Ma, S. Luo, P. Zeng, Y. Wei, Distributed IIoT anomaly detection scheme based on blockchain and federated learning, *J. Commun. Netw.* 26 (2) (2024) 252–262, <http://dx.doi.org/10.23919/JCN.2024.000016>.
- [13] L. Junwei, S. Muhammad, Y. Geng, J. Kai, C. Tie, M. Maode, Enhanced collaborative intrusion detection for industrial cyber-physical systems using permissioned blockchain and decentralized federated learning networks, *Eng. Appl. Artif. Intell.* 135 (2024) 108862, <http://dx.doi.org/10.1016/j.engappai.2024.108862>.
- [14] A. Govindaram, J. A. Enhancing industrial IoT security: Utilizing blockchain-assisted deep federated learning for collaborative intrusion detection, *J. Electr. Systems; Paris* 20 (2024) 1345–1363.
- [15] S. Mohanad, W.L. Wai, L. Siamak, P. Marius, HBFL: A hierarchical blockchain-based federated learning framework for collaborative IoT intrusion detection, *Comput. Electr. Eng.* 103 (2022) 108379, <http://dx.doi.org/10.1016/j.compeleceng.2022.108379>.
- [16] O. Bouachir, M. Aloqaily, Ö. Özkasap, F. Ali, Federatedgrids: Federated learning and blockchain-assisted p2p energy sharing, *IEEE Trans. Green Commun. Netw.* 6 (1) (2022) 424–436.
- [17] S. Singh, S. Rathore, O. Alfarraj, A. Tolba, B. Yoon, A framework for privacy-preservation of IoT healthcare data using Federated learning and blockchain technology, *Future Gener. Comput. Syst.* 129 (2022) 380–388.
- [18] M.H. Bhavsar, Y.B. Bekele, K. Roy, J.C. Kelly, D. Limbrick, FI-ids: Federated learning-based intrusion detection system using edge devices for transportation iot, *IEEE Access* 12 (2024) 52215–52226.
- [19] T. Deng, Y. Li, X. Liu, L. Wang, Federated learning-based collaborative manufacturing for complex parts, *J. Intell. Manuf.* 34 (7) (2023) 3025–3038.
- [20] W. Zhao, I.M. Aldyafiah, P. Gangwani, S. Joshi, H. Upadhyay, L. Lagos, A blockchain-facilitated secure sensing data processing and logging system, *IEEE Access* 11 (2023) 21712–21728.
- [21] T.K. Dasaklis, G.T. Voutsinas, F. Casino, A systematic literature review of the blockchain-enabled supply chain traceability implementations, *Sustainability* 14 (4) (2022) 2439.
- [22] K. Almutairi, S.J. Hosseini Dehshiri, S.S. Hosseini Dehshiri, A.X. Hoa, J. Arockia Dhanraj, A. Mostafaeipour, A. Issakhov, K. Techato, Blockchain technology application challenges in renewable energy supply chain management, *Environ. Sci. Pollut. Res.* 30 (28) (2023) 72041–72058.
- [23] X. Tang, L. Zhu, M. Shen, J. Peng, J. Kang, D. Niyato, A.A. Abd El-Latif, Secure and trusted collaborative learning based on blockchain for artificial intelligence of things, *IEEE Wirel. Commun.* 29 (3) (2022) 14–22.
- [24] S. Ali, Q. Li, A. Yousafzai, Blockchain and federated learning-based intrusion detection approaches for edge-enabled industrial IoT networks: A survey, *Ad Hoc Netw.* 152 (2024) 103320.
- [25] A. Belenguer, J.A. Pascual, J. Navaridas, A review of federated learning applications in intrusion detection systems, *Comput. Netw.* (2025) 111023.
- [26] S.O. Ajakwe, K.L. Olabisi, D.-S. Kim, Multihop intruder node detection scheme (MINDS) for secured drones' FANET communication, *IET Intell. Transp. Syst.* 19 (1) (2025) e70080.
- [27] F. Othmane, A.F. Mohamed, B. Mohamed, B. Tarek, K. Burak, R.C. Kim-Kwang, 2DF-IDS: Decentralized and differentially private federated learning-based intrusion detection system for industrial IoT, *Comput. Secur.* 127 (2023) 103097, <http://dx.doi.org/10.1016/j.cose.2023.103097>.
- [28] Y. Shan, Y. Yao, X. Zhou, T. Zhao, B. Hu, L. Wang, CFL-IDS: An effective clustered federated learning framework for industrial Internet of Things intrusion detection, *IEEE Internet Things J.* 11 (6) (2024) 10007–10019, <http://dx.doi.org/10.1109/JIOT.2023.3324302>.
- [29] S.O. Ajakwe, D.-S. Kim, EQAI: Explainable quantum-empowered antispoofing intelligence for trustworthy connected autonomous vehicles communication, *IEEE Internet Things J.* (2025).
- [30] V. Sugumaran, A. Rajaram, Lightweight blockchain-assisted intrusion detection system in energy efficient MANETs, *J. Intell. Fuzzy Systems* 45 (1) (2023) <http://dx.doi.org/10.3233/JIFS-231340>.
- [31] K. Salaheddine, B. Tajeddine, A. Nabih, A. Mounir, Multi-zone-wise blockchain based intrusion detection and prevention system for IoT environment, *J. Comput. Mater. Contin.* 74 (1) (2023) <http://dx.doi.org/10.32604/cmc.2023.032220>.
- [32] I.H. Abdulqadder, S. Zhou, I.T. Aziz, D. Zou, X. Deng, S.M. Abrar Akber, An effective lightweight intrusion detection system with blockchain to mitigate attacks in SDN/NFV enabled cloud, in: 2021 6th International Conference for Convergence in Technology (I2CT), 2021, pp. 1–8, <http://dx.doi.org/10.1109/I2CT51068.2021.9417961>.
- [33] J. Kaur, G. Singh, A blockchain-based machine learning intrusion detection system for Internet of Things, in: K. Daimi, I. Dionysiou, N. El Madhoun (Eds.), *Principles and Practice of Blockchains*, Springer International Publishing, Cham, 2023, pp. 119–134, http://dx.doi.org/10.1007/978-3-031-10507-4_6.
- [34] D. Lipsa, B. Vimal, L. Rajasvaran, D. Khushi, S. Bhuvi, AI model for blockchain based industrial IoT and big data, *Springer Int. Publ.* 6 (2023) 55–81, http://dx.doi.org/10.1007/978-3-031-31952-5_3.
- [35] S. Shitharth, S. Gautam, K. Alaa O., K. Adil O., B. Mohamed, A. Ali, C.-W.L. Jerry, An artificial intelligence lightweight blockchain security model for security and privacy in IIoT systems, *J. Cloud Comput.* 12 (38) (2023) <http://dx.doi.org/10.1186/s13677-023-00412-y>.

- [36] X. Tang, L. Zhu, M. Shen, J. Peng, J. Kang, D. Niyato, A.A.A. El-Latif, Secure and trusted collaborative learning based on blockchain for artificial intelligence of things, *IEEE Wirel. Commun.* 29 (3) (2022) 14–22, <http://dx.doi.org/10.1109/MWC.003.2100598>.
- [37] S.O. Ajakwe, D.-S. Kim, Federated Learning and Lightweight Blockchain for Resilient UAV Communication Against PNT and Model Poisoning Attacks.
- [38] L. Abdullah, A.M. Mazin, K. Seifedine, A.A. Salman, S.M. Mashael, H.A. Karar, Federated Learning-Aware Multi-Objective Modeling and blockchain-enable system for IIoT applications, *Comput. Electr. Eng.* 100 (2022) 107839, <http://dx.doi.org/10.1016/j.compeleceng.2022.107839>.
- [39] C.A. Nnadike, S.O. Ajakwe, J.-M. Lee, D.-S. Kim, RemoteCare: AI-driven multimodal predictive framework with blockchain for personalized remote patient monitoring in IoMT, *IEEE Internet Things J.* 13 (3) (2025) 4508–4523.
- [40] A. Samy, H. Yu, H. Zhang, Fog-based attack detection framework for Internet of Things using deep learning, *IEEE Access* 8 (2020) 74571–74585, <http://dx.doi.org/10.1109/ACCESS.2020.2988854>.
- [41] S. Alireza, N. Monire, A. Yousef, A new cloud-based cyber-attack detection architecture for hyper-automation process in industrial Internet of Things, *IEEE Access* 27 (2024) 3639–3655, <http://dx.doi.org/10.1007/s10586-023-04163-y>.
- [42] Q. Li, Z. Wen, Z. Wu, S. Hu, N. Wang, Y. Li, X. Liu, B. He, A survey on federated learning systems: Vision, hype and reality for data privacy and protection, *IEEE Trans. Knowl. Data Eng.* 35 (4) (2023) 3347–3366, <http://dx.doi.org/10.1109/TKDE.2021.3124599>.
- [43] L. D'hooge, The Canadian Institute for Cybersecurity datasets for network intrusion detection, 2023, URL: kaggle.com/datasets/dhoogla/cicidscollection.
- [44] M.K. Dahouda, I. Joe, A deep-learned embedding technique for categorical features encoding, *IEEE Access* 9 (2021) 114381–114391, <http://dx.doi.org/10.1109/ACCESS.2021.3104357>.
- [45] A. Chimeremma, Internal IIoT dataset generation script for PureChain lightweight FL-BC IDS, 2025, https://github.com/CHYMEE/lightweighted-Paper/blob/main/generate_internal_iiot_dataset.py.
- [46] I.S. Igboanusi, A. Allwinnaldo, R.N. Alief, M.R.R. Ansori, J.-M. Lee, D.-S. Kim, Smart auto mining (SAM) for industrial IoT blockchain network, *IET Commun.* 16 (18) (2022) 2123–2132, <http://dx.doi.org/10.1049/cmu2.12465>.
- [47] S.O. Ajakwe, I.S. Igboanusi, J.-M. Lee, D.-S. Kim, I BANDA: A blockchain-assisted defense system for authentication in drone-based logistics, *Drones* 9 (8) (2025) 590.
- [48] Networked Systems Laboratory, PureChain ecosystem: Proof of authority with association and smart auto mining, 2023, White paper, June 13.
- [49] D. Beutel, T. Topal, A. Mathur, X. Qiu, T. Parcollet, N. Lane, Flower: A friendly federated learning research framework, 2020.
- [50] G.B. Bhavana, et al., Comparative evaluation and simulation of blockchain consensus mechanisms, *Sci. Rep.* (2025) <http://dx.doi.org/10.1038/s41598-025-27431-w>.
- [51] R. Asif, S.R. Hassan, Shaping the future of Ethereum: Exploring energy consumption in Proof-of-Work and Proof-of-Stake consensus, *Front. Blockchain* 6 (2023) 1151724.
- [52] H. Kurlisaka, et al., Performance evaluation of ethereum consensus mechanisms for IoT systems using resource-constrained devices, *Clust. Comput.* (2025) <http://dx.doi.org/10.1007/s10586-025-05503-w>.
- [53] S.O. Ajakwe, I.I. Saviour, J.-H. Kim, D.-S. Kim, J.M. Lee, BANDA: A novel blockchain-assisted network for drone authentication, in: 2023 Fourteenth International Conference on Ubiquitous and Future Networks, ICUFN, IEEE, 2023, pp. 120–125.